Abstract Construction of Ideal Theory in Algebraic Number and Function Fields

Continuation of §6. Ideal theory under the divisor-chain condition

To obtain a representation by greatest primary components it is only necessary to replace, wherever possible, the representation by irreducible ideals whose existence is supplied by Theorem I - and which by Theorem II are primary - by a shortest such representation. If the primary ideals belonging to the same prime ideal are then collected together, the desired representation follows from §5, 3. The primary components may be called greatest components, since the least common multiple of any several of them is no longer primary.

§7. Ideal theory under the double chain condition

The simplifications, compared with the theory developed so far, rest on the two auxiliary results stated below.

First, if the multiple-chain condition holds in a commutative ring without zero divisors, then the ring is already a field. It is enough to show that, for $a \neq 0$, the equation

$$ax = b$$

always has a solution in the ring. It cannot have more than one solution, because the ring has no zero divisors. Let $\mathfrak{a}$ be the principal ideal generated by a. By hypothesis the sequence of powers of $\mathfrak{a}$ eventually stops; say $\mathfrak{a}^n$ is equal to all later powers. If $\mathfrak{b}$ denotes the principal ideal generated by b, then

$$\mathfrak{a}^n\mathfrak{b} = \mathfrak{a}^{n+1}\mathfrak{b}.$$

In particular, the element $a^n b$ has a representation of the form

$$a^n b = ka^{n+1}b + na^{n+1}b = a^{n+1}c,$$

where k, n are ordinary integers and c is again an element of the ring. Since zero divisors do not occur, this gives $b = ac$. Thus the field property is proved.

The preceding argument immediately gives the following lemma. If the multiple-chain condition holds in a commutative ring, then a prime ideal has no proper divisor different from the zero ideal. Likewise, if only axiom II is assumed - the multiple-chain condition modulo every ideal different from the zero ideal - then every prime ideal different from the zero ideal has no proper divisor different from 0. Indeed, the residue class ring modulo a prime ideal is, by definition, a ring without zero divisors; by homomorphy it too satisfies the multiple-chain condition, and hence is a field. By the one-to-one correspondence between divisors of the prime ideal and ideals of the residue class ring, the prime ideal has no proper divisor different from 0.

Theorem IV. If the double chain condition holds in a commutative ring, then in every shortest representation of an ideal the primary components which do not belong to the unit ideal are uniquely determined.

Supplement. Under axioms I and II the same assertion holds for every ideal different from the zero ideal.

Let

$$\mathfrak{m} = [\mathfrak{q}, \mathfrak{q}_1, \ldots, \mathfrak{q}_r] = [\bar{\mathfrak{q}}, \bar{\mathfrak{q}}_1, \ldots, \bar{\mathfrak{q}}_s]$$

be two shortest representations of $\mathfrak{m}$ by greatest primary components, and let

$$0, \mathfrak{p}_1, \ldots, \mathfrak{p}_r \quad \text{and} \quad 0, \bar{\mathfrak{p}}_1, \ldots, \bar{\mathfrak{p}}_s$$

be the corresponding prime ideals. The component $\mathfrak{q}$, respectively $\bar{\mathfrak{q}}$, is omitted if no primary ideal belonging to 0 occurs. Since

$$0 \cdot \mathfrak{p}_1 \cdots \mathfrak{p}_r \equiv 0 \pmod{\mathfrak{m}},$$

each $\bar{\mathfrak{p}}_j$ must be contained in one of the $\mathfrak{p}_i$, and hence, by the lemma, must be equal to it. Conversely each $\mathfrak{p}_i$ must agree with one of the $\bar{\mathfrak{p}}_j$. Thus the prime ideals different from 0 coincide.

Next, for instance,

$$\mathfrak{q}\mathfrak{q}_2 \cdots \mathfrak{q}_r \equiv 0 \pmod{\mathfrak{q}_1},$$

and therefore $\bar{\mathfrak{q}}_1 \equiv 0 \pmod{\mathfrak{q}_1}$, since the other components are not divisible by $\mathfrak{p}_1$ and are therefore prime to $\mathfrak{q}_1$. The reverse divisibility is obtained in the same way; hence the primary component belonging to each non-zero prime ideal is unique. If a component belonging to 0 occurs, then, because the representation is shortest, the corresponding component occurs in the other representation as well; this gives the remaining uniqueness. The same proof also shows that every representation containing no primary component belonging to 0 is itself shortest.

If, in addition to the double chain condition, the existence of a unit element is assumed, Theorem IV sharpens as follows.

Theorem V. In a commutative ring with unit element satisfying the double chain condition, every ideal can be represented uniquely as a product of finitely many pairwise relatively prime primary ideals.

Supplement. Under axioms I, II, III, Theorem V holds for every ideal different from the zero ideal.

Because the unit element exists, $0^0 = \mathfrak{o}$; hence every primary ideal belonging to 0 is equal to 0. Such a component cannot occur in a shortest representation of a non-zero ideal. The primary components are therefore uniquely determined by Theorem IV. At the same time every representation becomes shortest after omitting the zero component. Distinct prime ideals are, by the lemma, relatively prime; hence by the calculation rules of §4, 4, the same is true of the corresponding primary components. The least common multiple is therefore equal to their product.

The same hypotheses also give the following lemma. In a commutative ring with unit element and double chain condition, the prime ideals which are prime to an ideal $\mathfrak{m}$ are precisely all prime ideals, except the unit ideal, which are not contained in $\mathfrak{m}$. The notions "prime to" and "relatively prime" coincide.

Under axioms I, II, III the ideal $\mathfrak{m}$ is to be assumed different from the zero ideal.

If $\mathfrak{p}$ is not contained in $\mathfrak{m}$, then it is different from all prime ideals belonging to $\mathfrak{m}$. Hence, by the lemma above, it is divisible by none of these prime ideals, and so is prime to each primary component and therefore to $\mathfrak{m}$. It is also relatively prime to each component and hence relatively prime to $\mathfrak{m}$.

If $\mathfrak{p} \neq 0$ is contained in $\mathfrak{m}$, then it must coincide with one of the prime ideals belonging to $\mathfrak{m}$. Suppose it belongs to the component $\mathfrak{q} = \mathfrak{q}_1$. Then $\mathfrak{q} : \mathfrak{p}$ is a proper divisor of $\mathfrak{q}$, since $\mathfrak{p}^{e-1} \equiv 0 \pmod{\mathfrak{q} : \mathfrak{p}}$ but not modulo $\mathfrak{q}$. At the same time $\mathfrak{q} : \mathfrak{p}$, as a divisor of $\mathfrak{p}^{e-1}$, is divisible only by one non-zero prime ideal and is therefore primary. By unique factorization the product

$$(\mathfrak{q} : \mathfrak{p})\mathfrak{q}_2 \cdots \mathfrak{q}_r,$$

which is divisible by $\mathfrak{m} : \mathfrak{p}$, is a proper divisor of $\mathfrak{m}$. Thus $\mathfrak{p}$ is not prime to $\mathfrak{m}$.

Consequently an ideal $\mathfrak{t} \neq \mathfrak{p}$ is prime to $\mathfrak{m}$ precisely when none of the prime ideals belonging to $\mathfrak{t}$ is contained in $\mathfrak{m}$; in that case $\mathfrak{t}$ is also relatively prime to $\mathfrak{m}$. Since, conversely, two relatively prime ideals are always mutually prime (§5, 5), the two notions coincide under the hypotheses above.

§8. Ideal theory under integral closedness in the quotient field

The ideal theory developed up to this point is now to be sharpened to the usual one by adjoining the last two axioms.

First lemma. Let R be a commutative ring without zero divisors and with unit element, and assume the divisor-chain condition in R (axioms I, III, IV). Then from

$$\mathfrak{a} = \mathfrak{a}\mathfrak{b}, \qquad \mathfrak{a} \neq (0),$$

it always follows that $\mathfrak{b} = \mathfrak{o}$. Thus

$$\mathfrak{c} \neq \mathfrak{c}\mathfrak{t}$$

for every ideal different from the zero and unit ideals.

The proof is the same as in the lemma of §1, 3, since $\mathfrak{a}\mathfrak{b}$ may be regarded as a finite module with respect to $\mathfrak{b}$. If $\alpha_1, \ldots, \alpha_n$ is a basis of the ideal $\mathfrak{a}$, which exists by axiom I, then from $\mathfrak{a} = \mathfrak{a}\mathfrak{b}$ one obtains a system of equations

$$\alpha_i = b_{i1}\alpha_1 + \cdots + b_{in}\alpha_n, \qquad b_{ij} \in \mathfrak{b}.$$

Since R has no zero divisors, the determinant $|b_{ij} - e_{ij}|$ is zero, where $e_{ij} = 0$ for $i \neq j$ and $e_{ii} = 1$.

The equation obtained from this determinant has the form

$$1 + b_1 + \cdots + b_n = 0, \qquad b_i \in \mathfrak{b},$$

and hence $1 \in \mathfrak{b}$, so $\mathfrak{b} = \mathfrak{o}$.

It remains to show that, once integral closedness in the quotient field is added to axioms I-IV (axiom V), every primary ideal different from zero is a power of its associated prime ideal. The uniqueness of the resulting representation

$$\mathfrak{m} = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_r^{e_r}$$

for all ideals different from zero has already been proved by Theorem V and the preceding lemma; at the same time those prime ideals have no proper divisors except the unit ideal. The proof will first be given for exponent two, using Dedekind's second consequence from §1, 4, and then in general by complete induction.

Lemma. Under axioms I-V there are no primary ideals of exponent two except $\mathfrak{q} = \mathfrak{p}^2$.

It must be shown that from

$$\mathfrak{p}^2 \equiv 0 \pmod{\mathfrak{q}}, \qquad \mathfrak{q} \equiv 0 \pmod{\mathfrak{p}}, \qquad \mathfrak{q} \not\equiv 0 \pmod{\mathfrak{p}^2}$$

one necessarily gets $\mathfrak{q} = \mathfrak{p}^2$. Choose

$$\mathfrak{c} \equiv 0 \pmod{\mathfrak{q}}, \qquad \mathfrak{c} \equiv 0 \pmod{\mathfrak{p}}, \qquad \mathfrak{c}ot \equiv 0 \pmod{\mathfrak{p}^2}.$$

Then by the lemma of §7, 3, the ideal $\mathfrak{c} : \mathfrak{p}$ is a proper divisor of $\mathfrak{c}$. Thus there is an element b such that

$$b\mathfrak{p} \equiv 0 \pmod{\mathfrak{c}} \equiv 0 \pmod{\mathfrak{q}}, \qquad bot \equiv 0 \pmod{\mathfrak{c}}.$$

Consequently $\gamma = b/c$ is not integral: it belongs to the quotient field but not to R.

We shall prove $b \equiv 0 \pmod{\mathfrak{p}}$; then, since $\mathfrak{q}$ is primary and belongs to $\mathfrak{p}$, it follows that $\mathfrak{p} \equiv 0 \pmod{\mathfrak{q}}$ and hence $\mathfrak{q} = \mathfrak{p}^2$. By Dedekind's second consequence there exist elements m, n of R with

$$\gamma = b/c = m/n$$

such that m^2/n is still not integral. Equivalently,

$$bn = mc, \qquad m \not\equiv 0 \pmod{nR}, \qquad m^2 ot \equiv 0 \pmod{nR}.$$

Multiplying $b\mathfrak{p} \equiv 0 \pmod{\mathfrak{c}}$ by n gives

$$mb\mathfrak{p} \equiv 0 \pmod{n\mathfrak{c}},$$

and hence $m\mathfrak{p} \equiv 0 \pmod{nR}$, because the ring has no zero divisors. From $m^2 \equiv 0 \pmod{\mathfrak{p}nR}$ and

$n \not\equiv 0 \pmod{\mathfrak{p}}$ it follows that $m \equiv 0 \pmod{\mathfrak{p}}$; here one again uses that $nR : \mathfrak{p}$ is a proper divisor of nR. The four relations

$$m \equiv 0 \pmod{\mathfrak{p}}, \quad n \not\equiv 0 \pmod{\mathfrak{p}}, \quad c \equiv 0 \pmod{\mathfrak{p}}, \quad cot \equiv 0 \pmod{\mathfrak{p}^2}, \quad bn = mc$$

then force $b \equiv 0 \pmod{\mathfrak{p}}$. Indeed, since $\mathfrak{p}^2$ is primary, first $mc \equiv 0 \pmod{\mathfrak{p}^2}$, and from $bn = mc$ one obtains $b \equiv 0 \pmod{\mathfrak{p}}$. This proves the lemma.

A second lemma removes the restriction on the exponent.

Lemma. Under axioms I-V there are no primary ideals of exponent ρ except $\mathfrak{q} = \mathfrak{p}^\rho$.

This follows from the preceding lemma without further use of the axioms. First one proves:

$$\text{if } \mathfrak{c} \equiv 0 \pmod{\mathfrak{p}}, \quad \mathfrak{c}ot \equiv 0 \pmod{\mathfrak{p}^2}, \quad \text{then} \quad \mathfrak{p}^\rho = [\mathfrak{c}^\rho, \mathfrak{p}^{\rho+1}]$$

for every ρ. For $\rho = 1$ this is just the exponent-two lemma. If

$$\mathfrak{p}^{\rho-1} = [\mathfrak{c}^{\rho-1}, \mathfrak{p}^\rho],$$

then multiplication by $\mathfrak{p}$ and the distributive law give

$$\mathfrak{p}^\rho = [\mathfrak{c}^\rho, \mathfrak{c}^{\rho-1}\mathfrak{p}^{\rho+1}, \mathfrak{p}^{\rho+1}] = [\mathfrak{c}^\rho, \mathfrak{p}^{\rho+1}].$$

The assertion can be put in the following equivalent form. If

$$\mathfrak{q} \equiv 0 \pmod{\mathfrak{p}^\rho}, \qquad \mathfrak{q}ot \equiv 0 \pmod{\mathfrak{p}^{\rho+1}},$$

and if

$$\mathfrak{p}^{\rho+k} \equiv 0 \pmod{\mathfrak{q}}, \qquad k \geq 1,$$

then already

$$\mathfrak{p}^{\rho+k-1} \equiv 0 \pmod{\mathfrak{q}}.$$

Indeed, for every primary ideal there is an exponent $\rho > 1$ with $\mathfrak{q} \equiv 0 \pmod{\mathfrak{p}^\rho}$ but not modulo $\mathfrak{p}^{\rho+1}$; and $\mathfrak{q}ot \equiv 0 \pmod{\mathfrak{p}^{\rho+1}}$ follows from $\mathfrak{p}^\rho e \mathfrak{p}^{\rho+1}$ by the first lemma of this section. Repeated application of the second formulation gives $\mathfrak{p}^\rho \equiv 0 \pmod{\mathfrak{q}}$ and hence $\mathfrak{q} = \mathfrak{p}^\rho$.

From the hypotheses of the second formulation, and from the above representation of $\mathfrak{p}^\rho$, one obtains for every element g of $\mathfrak{q}$ a representation

$$g = bc^\rho \pmod{\mathfrak{p}^{\rho+1}}$$

with either $b \equiv 0 \pmod{\mathfrak{p}}$ or

$$bc^\rho \equiv 0 \pmod{[\mathfrak{q}, \mathfrak{p}^{\rho+k}]}.$$

Since $[\mathfrak{q}, \mathfrak{p}^{\rho+1}]$ is primary, it follows in the latter case that

$$c^\rho \equiv 0 \pmod{[\mathfrak{q}, \mathfrak{p}^{\rho+1}]},$$

or else

$$c^{\rho+k-1} \equiv 0 \pmod{[\mathfrak{q}, \mathfrak{p}^{\rho+k}]}.$$

In either case one obtains

$$\mathfrak{p}^{\rho+k-1} \equiv 0 \pmod{\mathfrak{q}}.$$

This proves the general lemma.

Combining the results gives:

Theorem VI. If axioms I-V hold in a commutative ring, then every ideal different from the zero and unit ideals is uniquely representable as a product of finitely many prime ideals different from the zero and unit ideals. These prime ideals have no proper divisors except the unit ideal.

§9. The axioms as consequences of the assumed decomposition

We now prove conversely that the usual ideal factorization implies the axioms. It is necessary to assume the factorization in the following sharpened form.

Assumption. In the commutative ring R, every ideal not generated by the zero or unit element is uniquely representable as a product of powers of prime ideals. These prime ideals are simple ideals not generated by the unit element; that is, they have no proper divisor different from 0. Conversely, all simple ideals are prime ideals. The uniqueness is to be understood sharply: for every ideal not generated by the zero or unit element,

$$\mathfrak{a}^\rho \neq \mathfrak{a}^{\rho+1}.$$

The existence of a unit element is not presupposed. If no unit element exists, the condition "not generated by the unit element" imposes no restriction.

First we prove axiom III, the existence of a unit element. Suppose axiom III fails. We shall show that R^2 is then a simple ideal without being a prime ideal, contradicting the assumption. By the uniqueness assumption $R \neq R^2$; hence

$$R \equiv 0 \pmod{R^2}, \qquad R^2 ot \equiv 0 \pmod{R^2},$$

and so R^2 is not a prime ideal. But R^2 is simple. Indeed, it can be divisible by no non-zero prime ideal, and therefore, by the assumed product representation, it has no divisors except 0 and R^2. This proves the existence of a unit.

Next comes axiom IV, absence of zero divisors. If a is a zero divisor, so that

$$ab = 0, \qquad a \neq 0, \qquad b \neq 0,$$

then multiplication of the representations of the principal ideals generated by a and b gives

$$(0) = \mathfrak{p}_1^{\rho_1} \cdots \mathfrak{p}_r^{\rho_r},$$

where the $\mathfrak{p}_i$ are different from the zero and unit ideals. The representation may be assumed shortest in the sense that omission of any factor gives a proper divisor of zero; this need not be automatic, since no uniqueness assumption has been made for the zero ideal. If the representation contains only one prime ideal, then

$$(0) = \mathfrak{p}^{\rho} = \mathfrak{p}^{\rho+1},$$

contrary to the sharpened uniqueness condition. If it contains more than one prime ideal, then because the unit element exists $\mathfrak{p}_1$ is relatively prime to all the other prime ideals; hence again the sharpened uniqueness condition is contradicted. Thus zero divisors do not occur.

The chain conditions, axioms I and II, follow as well. The assumptions imply that for every non-zero ideal, divisibility is equivalent to the occurrence of a product representation. More precisely, from

$$\mathfrak{m} \equiv 0 \pmod{\mathfrak{a}}, \qquad \mathfrak{a} e 0,$$

it follows that

$$\mathfrak{m} = \mathfrak{a}\mathfrak{b}, \qquad \mathfrak{b} \equiv 0 \pmod{\mathfrak{m}}.$$

If

$$\mathfrak{m} = \mathfrak{p}_1^{\rho_1} \cdots \mathfrak{p}_r^{\rho_r}, \qquad \mathfrak{a} = \mathfrak{q}_1^{\sigma_1} \cdots \mathfrak{q}_s^{\sigma_s},$$

then each $\mathfrak{q}_j$ must be identical with one of the $\mathfrak{p}_i$, and the corresponding exponent must not exceed ρ_i. Hence the divisors of $\mathfrak{m}$ are only the finitely many ideals corresponding to the combinations

$$0 \leq \sigma_i \leq \rho_i.$$

The residue class ring modulo $\mathfrak{m}$ therefore satisfies the double chain condition. This proves axioms I and II; the divisor-chain condition holds in R itself, since every proper divisor of the zero ideal is different from the zero ideal.

Before proving axiom V, integral closedness in the quotient field, we derive the usual consequences of ideal factorization.

First, the residue class ring modulo every non-zero ideal is a principal ideal ring. The ideal may be assumed different from the unit ideal, since for the residue ring consisting only of the zero element the assertion is clear. If the ideal is primary, say $\mathfrak{q} = \mathfrak{p}^{\rho}$, and if

$$\mathfrak{c} \equiv 0 \pmod{\mathfrak{p}}, \qquad \mathfrak{c} \not\equiv 0 \pmod{\mathfrak{p}^2},$$

then the argument of §8 gives

$$\mathfrak{p}^{\rho} = [\mathfrak{c}^{\rho}, \mathfrak{p}^{\rho+1}]$$

for every ρ. Thus in the residue class ring modulo a primary ideal every ideal is principal. In general, let

$$\mathfrak{m} = \mathfrak{q}_1 \cdots \mathfrak{q}_r$$

be the decomposition into pairwise relatively prime primary ideals, and let

$$R/\mathfrak{m} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_r$$

be the corresponding direct-sum decomposition. Each $\mathfrak{a}_i$ is isomorphic to $R/\mathfrak{q}_i$, and every ideal in $\mathfrak{a}_i$ is principal. If $\mathfrak{c}$ is an arbitrary ideal of the residue class ring, then $\mathfrak{a}_i\mathfrak{c}$ is a principal ideal, say $\mathfrak{a}_i c_i$. Setting

$$c = c_1 + \cdots + c_r$$

one obtains

$$\mathfrak{c} = Rc.$$

The ideal is therefore principal.

The principal-ideal-ring result may also be stated in the following form. If $\mathfrak{c}$ is any ideal, then it can be transformed into a principal ideal by multiplication with an ideal relatively prime to a prescribed ideal $\mathfrak{b}$. Indeed, set $\mathfrak{m} = \mathfrak{b}\mathfrak{c}$. Then $\mathfrak{c}$ becomes principal modulo $\mathfrak{m}$, i.e.

$$\mathfrak{c} = (cR, \mathfrak{m}) = cR + c\mathfrak{b} = c(R, \mathfrak{b}),$$

and hence $\mathfrak{c}\mathfrak{a} = cR$ with $(\mathfrak{a}, \mathfrak{b}) = R$.

We now pass to fractional ideals. A fractional ideal is by definition a finite R-module in the quotient field K.

Theorem. The non-zero finite R-modules in K form an abelian group under multiplication.

The product of two finite R-modules is again finite; multiplication is associative and commutative; and, since $R = R \cdot R$, the unit ideal is the identity element. It remains only to show that the equation

$$\mathfrak{A}X = R$$

has a solution in the system of finite R-modules.

We use the following preliminary remark. If the equation

$$\mathfrak{A}T = \mathfrak{B}$$

has one and only one solution, then

$$T = \mathfrak{B} : \mathfrak{A}.$$

For $T \equiv 0 \pmod{\mathfrak{B} : \mathfrak{A}}$, and conversely

$$\mathfrak{A}(\mathfrak{B} : \mathfrak{A}) = \mathfrak{B}.$$

If first $\mathfrak{A}$ is a principal module, $\mathfrak{A} = (\alpha)$, then $X = (1/\alpha)$ solves $\mathfrak{A}X = R$; it is again a finite R-module. It follows that every finite R-module is the quotient of two ideals of R, which justifies the term fractional ideal. Indeed, if $\tau_1, \ldots, \tau_n$ is a module basis of T, write $\tau_i = t_i/\alpha$ and let $\mathfrak{t}$ be the ideal generated by the t_i in R. Then

$$\alpha T = \mathfrak{t},$$

so by the preceding remark

$$T = \mathfrak{t} : (\alpha).$$

The inverse of a general finite module can now be written down. Suppose

$$\mathfrak{A} = \mathfrak{a} : \mathfrak{c},$$

and choose $\mathfrak{b}$ so that $\mathfrak{ab}$ is a principal ideal, say (α). Then

$$\mathfrak{A}\mathfrak{c} = \mathfrak{a}, \qquad \mathfrak{A}\mathfrak{b}\mathfrak{c} = (\alpha),$$

and therefore

$$\mathfrak{A}\left(\mathfrak{bc} : (\alpha)\right) = R.$$

The module $\mathfrak{bc} : (\alpha)$ is, as the quotient of an ideal by a principal ideal, again a finite R-module. The group property is proved. It also follows that the quotient of any two ideals $\mathfrak{a} : \mathfrak{b}$, as the solution of $\mathfrak{b}X = \mathfrak{a}$, is a finite R-module.

The group property has the following consequences.

First, cancellation is possible:

$$T = AM : BM \quad \iff \quad T = A : B.$$

Indeed, from $TBM = TAM$ follows $TB = TA$, and conversely.

Second, every fractional ideal admits a representation

$$\mathfrak{C} = \mathfrak{a} : \mathfrak{b}$$

where $\mathfrak{a}$ and $\mathfrak{b}$ are relatively prime; under this condition $\mathfrak{a}$ and $\mathfrak{b}$ are uniquely determined. Existence follows from the previous paragraph and cancellation. If

$$\mathfrak{C} = \mathfrak{a} : \mathfrak{b} = \bar{\mathfrak{a}} : \bar{\mathfrak{b}},$$

then $\mathfrak{a}\bar{\mathfrak{b}} = \mathfrak{b}\bar{\mathfrak{a}}$; uniqueness follows when both pairs are assumed relatively prime.

Third, every principal module generated by a non-integral element - an element belonging to the quotient field but not to R - has a representation as a quotient of principal ideals such that no power of the numerator is divisible by the denominator. Let

$$\eta R = \mathfrak{b} : \mathfrak{c}$$

be the reduced representation, so $(\mathfrak{b}, \mathfrak{c}) = R$. Choose, by the principal-ideal statement above, $\mathfrak{a}$ such that

$$\mathfrak{b}\mathfrak{a} = (b), \qquad (\mathfrak{a}, \mathfrak{c}) = R.$$

Then

$$\eta R = \mathfrak{a}\mathfrak{b} : \mathfrak{a}\mathfrak{c} = (b) : (c).$$

No power of b is divisible by c, because $(\mathfrak{a}, \mathfrak{c}) = R$ and $(\mathfrak{b}, \mathfrak{c}) = R$.

Now axiom V follows. Every non-integral element η of the quotient field admits a quotient representation

$$\eta = a/c$$

such that no power of the numerator is divisible by the denominator. Indeed, from $\eta R = (b) : (c)$ it follows that η differs from b/c only by a unit of R. Such an element can satisfy no equation characterizing integrality over R; for from

$$\eta^n + r_1\eta^{n-1} + \cdots + r_n = 0$$

one obtains

$$a^n \equiv 0yal \pmod{cR},$$

contrary to the construction. Thus R is integrally closed in its quotient field.

A useful corollary is the following. If axioms I-IV hold in a commutative ring and every primary ideal is irreducible, then axiom V also holds. Indeed, by axioms I-III every power $\mathfrak{p}^\rho$ of a prime ideal is primary, in particular $\mathfrak{p}^2$ is primary. By assumption $\mathfrak{p}^2$ is irreducible. It remains to prove a representation

$$\mathfrak{p} = (cR, \mathfrak{p}^2),$$

for then §8, 3 shows that every primary ideal is a power of its prime ideal, and this, together with the remaining hypotheses, implies integral closedness by the preceding paragraph.

By the divisor-chain condition one can write

$$\mathfrak{p} = (c_1R, \ldots, c_kR),$$

where the c_i are chosen linearly independent over the residue class field modulo $\mathfrak{p}$. If $k > 1$, this independence gives a decomposition

$$\mathfrak{p}^2 = [\mathfrak{q}_1, \mathfrak{q}_2],$$

where

$$\mathfrak{q}_1 = (c_1R, \mathfrak{p}^2), \qquad \mathfrak{q}_2 = (c_2R, \ldots, c_kR, \mathfrak{p}^2).$$

Both $\mathfrak{q}_1$ and $\mathfrak{q}_2$ are proper divisors of $\mathfrak{p}^2$, contradicting irreducibility. Therefore $k = 1$ and the required representation follows.

Conversely, under axioms I-V every primary ideal is irreducible. A power $\mathfrak{p}^\rho$ of a prime ideal cannot be written as the least common multiple of proper divisors of the form $\mathfrak{p}^\alpha$ and $\mathfrak{p}^\beta$.

If zero divisors are allowed in the ring, axioms I-III together with integral closedness in the quotient ring do not imply that every primary ideal is irreducible. Let T be a ring in which all axioms I-IV except integral closedness hold; such rings exist, for example finite orders different from the full ring of integral elements in a finite extension of the quotient field of a ring satisfying I-V. By the preceding corollary there is in T a reducible primary ideal $\mathfrak{q}$. Let $\mathfrak{p}^\rho$ be a proper multiple of $\mathfrak{q}$, and let

$$R = T/\mathfrak{p}^\rho$$

be the residue class ring. Then the ideal corresponding to $\mathfrak{q}$ becomes in R a non-zero reducible primary ideal. The ring R satisfies axioms I-III and is nevertheless integrally closed in its quotient ring. For every element of T not divisible by $\mathfrak{p}$ becomes relatively prime to $\mathfrak{p}^\rho$, so in R every regular element is a unit; hence R coincides with its quotient ring.

§10. Double chain condition and composition series

We shall now show that, for arbitrary well-ordered module domains (§2), the validity of the double chain condition - every divisor chain and every multiple chain of modules terminates - is equivalent to the existence of a composition series. Specializing the module domain to a commutative ring then gives the corresponding facts for the system of all ideals of the ring. In that case "module isomorphism" below is to be replaced throughout by "ring isomorphism".

A module domain G is called **simple** if there are in G no R-modules other than G itself and the zero module 0. A module A is called **simple in** G if G/A is simple.

A multiple chain

$$G, U_1, \ldots, U_r, 0$$

is called a composition series of G of length r if all modules in the chain are distinct and each module is simple in the preceding one. Thus the residue modules, or quotient groups,

$$U_{i-1}/U_i$$

are simple module domains, and similarly G/U_1 and $U_r/0$. By forming the residue module G/A the case of a composition series from G down to a module $A \neq 0$ is reduced to the absolute case.

Assume first that the double chain condition holds in G. Then there exists a composition series in G, provided $G \neq 0$. From the assumed well-ordering of G and the divisor-chain condition, a well-ordering of the system of all modules is obtained by the quasi-lexicographic arrangement used in §6. By this well-ordering and the divisor-chain condition, at least one module simple in G exists.

Let G_1 be the first proper divisor of G in the well-ordering, and, in general, let G_i be the first proper divisor of G_{i-1}. The well-defined chain

$$G, G_1, G_2, \ldots$$

terminates by the divisor-chain condition and so necessarily reaches a module A simple in G. If $A \neq 0$, the same procedure gives a module simple in A, and so on. We obtain a well-defined multiple chain which, by the multiple-chain condition, must terminate. It is therefore a composition series of G.

Conversely, suppose that a composition series exists in G. Then the double chain condition holds in G. The proof proceeds by complete induction, and no well-ordering of G is needed for this direction. During the induction one also proves the Jordan-Hölder theorem under the sole assumption that a composition series exists.

If G is simple, so that the only composition series is $G, 0$, the following assertions are immediate:

(a) Through every module simple in G one can draw a composition series.

(b) Jordan-Hölder theorem: every composition series has the same length and the same system of quotient groups, up to order and isomorphism.

(c) Through every module of G different from 0 one can draw a composition series.

(d) The multiple-chain condition holds.

(e) The divisor-chain condition holds.

Assume, then, that the assertions (a)-(e) have been proved for every module domain possessing a composition series of length $r < r+1$. We prove them for a module domain G with a composition series

$$G, U, U_1, \ldots, U_r, 0$$

of length $r+1$.

First, if there is no module simple in G other than U, there is nothing to prove. Let B be a module simple in G and different from U. Since U and B are both simple in G and are distinct, necessarily

$$(U, B) = G.$$

Put

$$M = [U, B].$$

The second isomorphism theorem (§4, 2) gives

$$G/B \simeq U/M, \qquad G/U \simeq B/M.$$

Thus M is simple in both U and B. Since U has a composition series of length $r < r+1$, the induction hypotheses (a) and (d) give a composition series in U passing through M, say

$$U, M, M_1, \ldots, M_{r-1}, 0.$$

Then

$$G, B, M, M_1, \ldots, M_{r-1}, 0$$

is a composition series of G passing through B.

To prove the Jordan-Hölder theorem, let

$$G, A, A_1, \ldots, A_s, 0 \quad \text{and} \quad G, B, B_1, \ldots, B_t, 0$$

be two composition series of G. If $A = B$, the result follows immediately from the induction hypothesis for length $r < r+1$. If $A \neq B$, comparison with the series constructed above,

$$G, A, M, M_1, \ldots, M_{r-1}, 0 \quad \text{and} \quad G, B, M, M_1, \ldots, M_{r-1}, 0,$$

shows the following. By the induction hypothesis the systems of quotient groups of the first given series and of the first constructed series agree; likewise for the second given series and the second constructed series, since the latter passes through B and has length $r+1$. Hence the lengths are equal. Moreover, the isomorphism noted above gives the required agreement between the quotients arising in the two constructed series. Therefore the two original composition series have the same quotient groups up to order and isomorphism. This proves the Jordan-Hölder theorem.

We next need a preliminary observation. Let A, B be modules in G and let B be simple in A. Then the modules (B, H) and (A, H) are either equal, or (B, H) is simple in (A, H). Indeed, by the second isomorphism theorem,

$$(A, B, H)/(B, H) \simeq A/[A, (B, H)].$$

Since $B = 0(A)$, this may also be written with

$$C = [A, (B, H)].$$

Now C lies between A and B, and hence, by the simplicity assumption, is equal either to A or to B. The assertion follows.

Let

$$G, U, U_1, \ldots, U_r, 0$$

again be a composition series of G, and let H be any module of G different from 0. Form the series

$$G, (U, H), (U_1, H), \ldots, (U_r, H), H.$$

By the preliminary observation, the distinct modules among these form a composition series from G down to H. Therefore G/H has a composition series, and hence, by the induction hypothesis, one can draw a composition series through H in G/H. Adding H gives a composition series of G through H. Repetition shows moreover that such a series may be taken as an extension of the one just constructed from G down to H.

The multiple-chain condition is now immediate. Suppose again that G has a composition series of length $r+1$, and let

$$G, B, B_1, \ldots, B_n, \ldots$$

be a multiple chain, each module being a proper multiple of the preceding one. There is no loss in beginning the chain with G, since G may always be adjoined as first term. By the preceding paragraph there is a composition series through B in G, and therefore B has a composition series of shorter length. By the induction hypothesis the chain

$$B, B_1, \ldots, B_n, \ldots$$

terminates; hence the chain with G adjoined terminates as well.

Finally, the divisor-chain condition is proved similarly. Let

$$G, G_1, G_2, \ldots, G_n, \ldots$$

be a divisor chain in which every module is a proper divisor of the preceding one. Again it is no restriction that the chain begins with G. By the preceding paragraph there is a composition series through G_1 in G; hence G/G_1 has a composition series of shorter length. By the induction hypothesis the chain

$$G/G_1, \quad G_2/G_1, \quad \ldots$$

terminates in G/G_1. By the one-to-one correspondence supplied by the first isomorphism theorem (§4, 2), the original chain beginning with G_1 also terminates; therefore the chain with G adjoined terminates. The equivalence of the two assumptions - existence of a composition series and the double chain condition - is proved.

The Discriminant Theorem for the Orders of an Algebraic Number or Function Field

The Dedekind discriminant theorem says, as is well known, that a prime number divides the field discriminant if and only if it is divisible by at least the square of a prime ideal.

I show here that this theorem remains valid for all finite orders of a finite algebraic number field - that is, for all rings of algebraic integers which contain the ring of rational integers - in the following form: a prime number p divides the discriminant of an order if and only if, in the ideal decomposition valid in that order of the principal ideal (p) into greatest primary components, at least one proper primary ideal occurs, that is, a primary ideal which is not itself a prime ideal. Since in the principal order - the system of all algebraic integers of the field - there are no primary ideals other than powers of prime ideals, Dedekind's theorem follows as a special case.

The proof is obtained by passing from the order modulo (p) to the residue class ring. The assertion is thereby reduced to the decomposition theory of rings of finite rank over a field contained in the ring. Such a ring is a commutative system of hypercomplex quantities with principal unit and with coefficients from that field. In the present case the coefficient field is the residue class field of the rational integers modulo p, and the decomposition of (p) corresponds in the residue class ring to the decomposition of the zero ideal.

Call such a finite-rank ring **completely reducible** if its zero ideal can be represented as the least common multiple of finitely many prime ideals. We are then concerned with criteria for complete reducibility. First, if the coefficient field is perfect, the ring is completely reducible if and only if the extension ring obtained by passing to an algebraically closed coefficient field is completely reducible. For this extension ring the non-vanishing of the discriminant is an especially simple necessary and sufficient criterion of complete reducibility. Since the discriminant of the order modulo every prime number passes into the discriminant of the residue class ring, the discriminant theorem follows.

For function fields in place of number fields the order must contain the ground domain of all integral rational functions; for algebraic functions in several indeterminates or for integral algebraic functions it must contain the ground domain of all integral functionals with integral-rational coefficients. In this situation the imperfect coefficient-field case may occur in the residue class ring, for example when, in the integral functional domain case, one passes to the residue class ring modulo a prime number. With an imperfect coefficient field, prime ideals of the first and second kind must be distinguished according as the residue class field modulo the prime ideal is an extension field of the first or second kind. Correspondingly, complete reducibility of the first and second kind must be distinguished. The criteria just stated concern only complete reducibility of the first kind, which alone occurs in the perfect case.

In its most general form, then, the discriminant theorem says: a prime element of the ground domain divides the discriminant of an order if and only if in the ideal decomposition of (p) either at least one proper primary component or a prime ideal of the second kind occurs.

For the principal order this fact was first observed in examples by Kronecker, after the earlier Festschrift statement had still contained, mistakenly, the form valid for the principal order of a number field, though without a complete proof. For the principal order Ostrowski gave a proof of this discriminant theorem and further ramification theorems connected with it, along with a detailed account of the literature. The discriminant theorem for all orders of relative fields follows directly by passing to the quotient ring with respect to each prime ideal of the ground domain. The idea of this passage, known in special cases, has been emphasized in principle by W. Krull; a systematic theory of quotient rings in a more general context is given by H. Grell.

The uniform treatment here of all orders, and of both perfect and imperfect coefficient fields, rests not only on ideal theory but also on taking from the start the theory of rings of finite rank, and hence the general concept of finite extension, as fundamental, rather than the concept of an extension generated by a primitive element. The module basis of an order, together with the relations among its elements, takes the place of the powers of a primitive element and its defining equation, or the place of the powers of a fundamental form and its fundamental equation.

Such a module basis passes modulo every prime element of the ground domain into a basis of the residue class ring. This need not be true, even for the principal order, of a basis consisting of powers of a primitive element; nor of a basis consisting of powers of the fundamental form, once integral algebraic functions of several indeterminates are involved. Equivalently, the residue class ring of the polynomial ring in one indeterminate, with coefficients in the ground domain, modulo an arbitrary defining equation - even the fundamental equation - need not be isomorphic modulo p to the residue class ring of the principal order modulo (p). The main difficulties in the usual representations lie precisely in this failure of isomorphism. Since the auxiliary polynomial ring does not occur here, these difficulties are automatically avoided. At the same time the additive decomposition theory of the residue class ring becomes the equivalent of the multiplicative decomposition of the equation; when the equation decomposes into relatively prime factors, a corresponding additive decomposition occurs in the polynomial residue class ring.

It should be added that the discriminant theorem can only be the first theorem of a general ramification theory for orders, just as for the principal order the discriminant theorem is accompanied by the finer theory of the different or ramification ideal, which there permits a more precise determination of exponents. This determination of exponents, resting on the power representation of the primary factors of equations, may be interpreted as determination of the length of a composition series, from p to $\mathfrak{q} = p^t$, and will presumably appear in this form in the general ramification theory.

§1. Extension rings of fields

We begin with a few remarks about arbitrary rings. If T is a ring and S is a system of elements of T, the "ring derived from S in T" means the intersection of all rings in T which contain S. Correspondingly, one defines the "ideal derived from S in T", or the "P-module derived from S in T" when P is a subring of T.

If R is a subring of T and S is a system of elements of T, the ring derived from R and S in T is denoted by

$$R[S]$$

and is called the ring obtained by ring-adjoining S to R. It consists of all polynomials in the elements of S with coefficients in R, where equality and the operations are fixed by the equality and operation relations already present in T. Because these relations have already been fixed, in special cases it is enough to form only a subsystem of all polynomials - for example, all linear forms in S with coefficients in R. Such a case occurs, for instance, when R is a ring with unit element and S is a field with the same unit element. Then all linear combinations

$$\sum c_i y_i,$$

with $c_i \in S$ and $y_i \in R$, form the ring $R[S]$.

One can also construct the extension ring $R[S]$ when only the ring R is given and, in addition, a system S of elements not lying in R is given with equality relations and perhaps mutual operation relations. One shows that there is always a ring containing R and S, namely the ring formally formed from all polynomials in S with coefficients in R, subject to the usual operation rules. Apart from these operation rules, there remains complete freedom in defining equality, except that the equality definitions already valid in R and S must be included. Thus one is forced to identify only those polynomials which, by equality in R and S and by the stated operations, are built in the same way from equal elements of R and S. The ring T so constructed, containing R and S, is then identical with the ring derived in T from R and S.

From now on, rings R are assumed to have a unit element and to contain a field P as subring; they are thus over-rings of fields. The unit element e of R is also the unit element of P, and R is a P-module.

If $\bar{P}$ is an overfield of P, the extension ring

$$\bar{R} = R[\bar{P}]$$

obtained by ring-adjoining $\bar{P}$ to R is to be understood as follows. Assume that the set-theoretic intersection of R and $\bar{P}$ is exactly P, and that no operation relations exist between elements of R not in P and elements of $\bar{P}$ not in P. This assumption can always be achieved by replacing $\bar{P}$ by an equivalent extension field of P, or R by an equivalent extension ring.

The elements of $\bar{R}$ are all formally formed linear combinations

$$c_1 y_1 + cdots + c_n y_n,$$

where the c_i run through the elements of $\bar{P}$ and the y_i through the elements of R. The definition is to be unambiguous: if the y_i are replaced by equal elements of R, and the c_i by equal elements of $\bar{P}$, the element of $\bar{R}$ is to be the same.

The sum is defined in the evident way,

$$\sum c_i y_i + \sum d_i y_i = \sum (c_i + d_i) y_i,$$

where zeros may occur among the c_i, d_i so that formally the same y_i occur. The product is likewise defined by distributivity,

$$\left(\sum c_i y_i\right)\left(\sum d_j y_j\right) = \sum c_i d_j y_i y_j.$$

For equality one declares the following. If

$$y_{i_1}, \ldots, y_{i_s}$$

are linearly independent over P, then two elements

$$\sum c_j y_{i_j}, \qquad \sum d_j y_{i_j}$$

are equal if and only if $c_j = d_j$ for all j in $\bar{P}$. In other words, elements of R linearly independent over P are declared linearly independent over $\bar{P}$.

This gives equality for all elements of $\bar{R}$, because every element can be expressed by linearly independent elements of R. Indeed, from a relation

$$u = c_1 y_1 + \cdots + c_n y_n$$

in $\bar{R}$, multiplication by $b = be$ gives

$$bu = bc_1 y_1 + \cdots + bc_n y_n,$$

so that all elements are expressible in terms of linearly independent elements of R. Equality is therefore coefficient equality over $\bar{P}$. For elements lying simultaneously in R and $\bar{R}$, or in P and $\bar{P}$, this agrees with the old equality. For the remaining elements of $\bar{R}$, the equality in R and $\bar{P}$ says no more than the required unambiguity of the definition.

Nor does the requirement that sum and product be unambiguous introduce further relations: equal elements may be assumed coefficientwise equal, and hence they give formally equal sums and products. One checks at once that all ring axioms hold. Thus the extension ring $R[\bar{P}]$ generated by ring adjunction always exists.

Now let $\mathfrak{a}$ be an ideal of R. The module product

$$\bar{R}\mathfrak{a}$$

is an ideal $\bar{\mathfrak{a}}$ of $\bar{R}$, the extension ideal of $\mathfrak{a}$. It is also the ideal derived from $\mathfrak{a}$ in $\bar{R}$; it consists of all finite linear combinations

$$\sum c_i a_i,$$

where $a_i \in \mathfrak{a}$ and $c_i \in \bar{P}$.

If $\bar{\mathfrak{b}}$ is an ideal of $\bar{R}$, then the intersection

$$[\bar{\mathfrak{b}}, R]$$

is an ideal $\mathfrak{b}$ of R, the contraction ideal of $\bar{\mathfrak{b}}$. Every ideal $\mathfrak{a}$ of R is a contraction ideal, namely the contraction of its own extension ideal. For an element a of $[R, \bar{\mathfrak{a}}]$ has the form $\sum c_i a_i$ with the a_i chosen linearly independent over P and hence over $\bar{P}$. It follows that the elements a, a_i are linearly dependent in $\bar{R}$, and therefore, by the equality definition, already in R; hence a belongs to $\mathfrak{a}$. The converse is immediate.

If $\bar{\mathfrak{p}}$ is a prime ideal of $\bar{R}$, then

$$\mathfrak{p} = [R, \bar{\mathfrak{p}}]$$

is a prime ideal of R. By the second isomorphism theorem, the residue ring $R/\mathfrak{p}$ is isomorphic to a subring of $\bar{R}/\bar{\mathfrak{p}}$, and hence has no zero divisors. More precisely,

$$(R, \bar{\mathfrak{p}})/\bar{\mathfrak{p}} \simeq R/[R, \bar{\mathfrak{p}}],$$

where $(R, \bar{\mathfrak{p}})$ denotes the sum, the greatest common divisor, of the modules R and $\bar{\mathfrak{p}}$.

§2. Rings of finite rank. Representation as a direct sum

If R is a finite P-module, then R has finite rank over P, say n: there are n, but no more, linearly independent elements over P, and every system of n linearly independent elements is a module basis. If a P-module B of finite rank is divisible by another P-module A of the same finite rank, then they are equal.

From now on R is assumed to have finite rank n over P. Then every P-module in R has finite rank. In every proper divisor or multiple chain of P-modules in R, the rank of each module must therefore be greater, respectively smaller, than the rank of the preceding module; hence the chains terminate. In particular the double chain condition holds for the system of all ideals of R.

It follows that a prime ideal $\mathfrak{p}$ of R has no proper divisor other than the unit ideal. Thus the residue ring $R/\mathfrak{p}$ modulo a prime ideal different from the unit ideal is a field. If $\mathfrak{q}$ is a primary ideal belonging to $\mathfrak{p}$, say

$$\mathfrak{q} \equiv 0 \pmod{\mathfrak{p}}, \qquad \mathfrak{p}^\rho \equiv 0 \pmod{\mathfrak{q}},$$

then $R/\mathfrak{q}$ is a primary ring: a power, in fact at latest the ρ-th power, of every zero divisor vanishes. Further, $R/\mathfrak{q}$ contains only zero divisors and units. This is immediate because every ideal not divisible by $\mathfrak{p}$ is relatively prime to $\mathfrak{p}^\rho$ and hence to $\mathfrak{q}$.

From the double chain condition and the existence of the unit element it follows that every ideal $\mathfrak{a}$ of R is uniquely representable as a product of finitely many pairwise relatively prime primary ideals. Because of relative primality, this product is the least common multiple, and it corresponds to a representation of the residue class ring $R/\mathfrak{a}$ as a direct sum of primary rings: $R/\mathfrak{a}$ is the greatest common divisor of these rings, and the sum representation of every element of $R/\mathfrak{a}$ is unique.

Let the decomposition of the zero ideal of R be

$$(0) = \mathfrak{q}_1\mathfrak{q}_2 \cdots \mathfrak{q}_r = [\mathfrak{q}_1, \mathfrak{q}_2, \ldots, \mathfrak{q}_r],$$

and let the corresponding direct-sum decomposition be

$$R = R_1 + \cdots + R_r, \qquad R_iR_j = (0) \quad (i \neq j), \qquad R_i^2 = R_i.$$

Here R_i is an ideal of R, and

$$R_i \simeq R/\mathfrak{q}_i;$$

the element $y_i \in R_i$ corresponds to the class represented by y_i in $R/\mathfrak{q}_i$.

If $\mathfrak{a}$ is any ideal of R, the distributive law gives

$$\mathfrak{a} = R_1\mathfrak{a} + \cdots + R_r\mathfrak{a}.$$

The components

$$\mathfrak{a}_i = R_i\mathfrak{a}$$

are ideals both in R_i and in R. As ideals they are finite P-modules, and because the sum is direct their ranks add to the rank of $\mathfrak{a}$ or of R.

If the unit element has the decomposition

$$e = e_1 + \cdots + e_r,$$

where

$$e_ie_j = 0 \quad (i \neq j), \qquad e_i^2 = e_i,$$

then every element y of R has the unique decomposition

$$y = y_1 + \cdots + y_r, \qquad y_i = ye_i \in R_i,$$

and $R_i = Re_i$. The orthogonality relations show that the i-th component of a product is y_iz_i, and the i-th component of a sum is $y_i + z_i$.

The ring R_i contains a subfield

$$P_i = Pe_i$$

isomorphic to P, and has finite rank over P_i; this rank is equal to the rank of R_i as P-module. For if $c \in P$, then $ce_i = ce$ modulo $\mathfrak{q}_i$, so $ce_i \neq 0$ whenever $c \neq 0$. Hence $P \to P_i$ is an isomorphism. Linear dependence over P among elements of R_i passes, by multiplication with e_i, to linear dependence over P_i, and conversely, since $P_i = Pe_i$, every relation over P_i is already a relation over P. Thus the ranks agree.

If

$$R = R_1 + \cdots + R_r,$$

then for any extension ring $\bar{R} = R[\bar{P}]$ there is a corresponding decomposition

$$\bar{R} = \bar{R}_1 + \cdots + \bar{R}_r,$$

where $\bar{R}_i$ denotes the extension ideal of R_i in $\bar{R}$, and at the same time the ring derived from R_i in $\bar{R}$. Indeed, from $(0) = \mathfrak{q}_1 \cdots \mathfrak{q}_r$ follows by multiplication with $\bar{R}$ the product representation

$$(0) = \bar{\mathfrak{q}}_1 \cdots \bar{\mathfrak{q}}_r,$$

and the $\bar{\mathfrak{q}}_i$ remain pairwise relatively prime. The direct summands are the corresponding products of the other factors, hence the extension ideals of the R_i. But this extension ideal is, by §1, 3, just the extension ring

$$R_i[P_i]$$

with $P_i = Pe_i$. Thus in extending one may extend the individual components separately. In general $\bar{R}_i$ need no longer be a primary ring.

We shall also need a distinction between prime ideals of first and second kind. Modulo every prime ideal $\mathfrak{p}$ different from the unit ideal, the residue class ring $R/\mathfrak{p}$ is a field. It contains a subfield isomorphic to P, namely the classes represented by elements of P. This follows from the second isomorphism theorem:

$$(P, \mathfrak{p})/\mathfrak{p} \simeq P/[P, \mathfrak{p}] \simeq P,$$

because $\mathfrak{p}$ contains no non-zero element of P. The field $R/\mathfrak{p}$ has finite rank over this copy of P, and is therefore a finite algebraic extension field. According as this extension is of the first or second kind, $\mathfrak{p}$ is called a prime ideal of the first or second kind.

§3. Completely reducible rings

The ring R is called **completely reducible** if in the product decomposition of its zero ideal all primary components are prime ideals. Equivalently, R can be represented as a direct sum of finitely many fields. The completely reducible ring is called completely reducible of the first kind if all its prime ideals are of the first kind; otherwise it is completely reducible of the second kind.

If P is a perfect field, only complete reducibility of the first kind can occur; this is in particular always the case when P is algebraically closed. In what follows $\bar{P}$ denotes the algebraic closure of P, and

$$\bar{R} = R[\bar{P}].$$

Theorem. The ring R is completely reducible of the first kind if and only if the extension ring

$$\bar{R} = R[\bar{P}]$$

with algebraically closed $\bar{P}$ is completely reducible.

The proof has three parts.

First, complete reducibility of an extension ring implies complete reducibility of R. More generally, if the zero ideal of an extension ring has a representation as a least common multiple of prime ideals,

$$(0) = [\bar{\mathfrak{p}}_1, \ldots, \bar{\mathfrak{p}}_r],$$

then by intersecting with R one obtains the representation

$$(0) = [R, \bar{\mathfrak{p}}_1], \ldots, [R, \bar{\mathfrak{p}}_r]$$

of the zero ideal in R; and the contracted ideals are prime by §1, 3.

Second, complete reducibility of the first kind of R implies complete reducibility of the first kind of every extension ring, and more generally of every intermediate ring between R and the extension ring. Since by §2, 3 the direct-sum decomposition of R induces one of any extension ring, it is enough to consider one component. Thus we may assume without loss of generality that R is a field, and indeed a finite extension field of the first kind of its subfield P. Then R is obtained from P by adjoining one primitive element of the first kind. Equivalently,

$$R \simeq P[x]/(f(x)),$$

where $f(x)$ is a prime polynomial of the first kind in the polynomial ring $P[x]$. If Q is any extension field of P, then

$$Q[x]/(f(x)) \simeq R[Q],$$

for the rank of the residue ring remains equal to the degree of $f(x)$ when passing from $P[x]$ to $Q[x]$, and hence this is exactly the construction of the extension ring described in §1, 2. In $Q[x]$ the polynomial $f(x)$ decomposes into pairwise relatively prime prime functions of the first kind; each generates a prime ideal in $Q[x]$. Therefore $Q[x]/(f(x))$, and hence $R[Q]$, is completely reducible of the first kind. Taking $Q = \bar{P}$ proves the second step.

Third, if R is completely reducible of the second kind, then $\bar{R}$ is not completely reducible. As before it suffices to take R to be a field, now a finite extension field of the second kind of P. We must show that in the direct-sum decomposition of $\bar{R}$ into primary rings at least one proper primary component occurs. It is enough to exhibit a non-zero element of $\bar{R}$ whose power is zero: then some component is non-zero but nilpotent, and the corresponding component ring is not a field.

Let γ be an element of the second kind in R, of exponent $\ell > 1$, and let

$$F(\gamma) = \gamma^{kp^\ell} + a_1\gamma^{kp^\ell-1} + \cdots + a_{kp^\ell} = 0$$

be a lowest-degree equation satisfied by γ over P, where p is the characteristic of P. The elements

$$1, \gamma, \gamma^2, \ldots, \gamma^{kp^\ell-1}$$

are linearly independent over P and therefore remain linearly independent over $\bar{P}$ by the equality definition. If

$$G(\gamma) = \gamma^{kp^{\ell-1}} + \cdots$$

is chosen so that

$$G(\gamma)^p = F(\gamma),$$

then $G(\gamma)$ is non-zero in $\bar{R}$ but its p-th power vanishes. This proves the third step.

The first and third steps show that complete reducibility of R of the first kind follows from complete reducibility of $\bar{R}$; the second step gives the converse.

As a consequence, if R is completely reducible of the first kind, then $\bar{R}$ is a direct sum of rings of rank one, hence of fields isomorphic to $\bar{P}$. Such a decomposition already exists in some ring $R[Q]$, where Q is a finite extension field of P.

Indeed, by the theorem, $\bar{R}$ is a direct sum of fields which, as finite extensions of subfields isomorphic to $\bar{P}$, are identical with those subfields. Thus

$$\bar{R} = \bar{P}e_1 + \cdots + \bar{P}e_n, \qquad e = e_1 + \cdots + e_n,$$

and the components e_i of the identity form a linearly independent module basis of $\bar{R}$. Expressing the e_i as linear combinations of elements of R with coefficients in $\bar{P}$, the set of all coefficients generates a finite extension field Q of P. The e_i therefore already belong to $R[Q]$, and the same decomposition exists there. In every intermediate ring between $R[Q]$ and R the indecomposable components arise as extension rings of the Qe_i.

§4. Matrix representation, traces, and discriminants in rings of finite rank

In this and the next section the hypotheses on the base ring are slightly more general than before, in order also to cover orders in number and function fields. The treatment is a general and uniform formulation of essentially known facts.

Assume that R is an extension ring of an integral domain Z, and that the unit element of R already lies in Z. For every Z-module in R, in particular for R itself, at least one finite module basis exists which consists of elements linearly independent over Z. Two such bases $a_1, \ldots, a_n$ and $b_1, \ldots, b_n$ are bases of the same Z-module if and only if the transformation determinant is a unit of Z. This hypothesis is plainly fulfilled for the finite-rank rings considered above, where Z is the field P, and it is also fulfilled for orders in number and function fields, where Z is the ring of rational integers, respectively the functional domain.

Let $a_1, \ldots, a_n$ be a linearly independent Z-module basis of an ideal $\mathfrak{a}$, and let y be an arbitrary element of R. Since ya_i again belongs to $\mathfrak{a}$, equations with coefficients in Z hold:

$$ya_i = c_{i1}a_1 + \cdots + c_{in}a_n.$$

In matrix form, if $(a_1, \ldots, a_n)$ denotes a row matrix,

$$y(a_1, \ldots, a_n) = (a_1, \ldots, a_n)C.$$

As y ranges over R, the matrices C form a ring $R_{\mathfrak{a}}$ homomorphic to R. More precisely,

$$R_{\mathfrak{a}} \simeq R/((0) : \mathfrak{a}),$$

where $(0) : \mathfrak{a}$ is the ideal quotient of the zero ideal by $\mathfrak{a}$. Differences and products of elements correspond to differences and products of matrices; scalar elements $c \in Z$ correspond to diagonal matrices cE. Exactly those elements y for which $y\mathfrak{a} = (0)$ correspond to the zero matrix, which proves the stated isomorphism.

A different basis of $\mathfrak{a}$ gives an equivalent matrix ring. If

$$(\bar{a}_1, \ldots, \bar{a}_n) = (a_1, \ldots, a_n)P$$

with determinant of P a unit of Z, then the corresponding matrices satisfy

$$\bar{C} = P^{-1}CP.$$

Two ideals $\mathfrak{a}$ and $\mathfrak{b}$ of R belong to the same ideal class if they are isomorphic as R-modules; that is, their elements correspond one-to-one in such a way that difference and multiplication by the same element of R correspond. Two matrix rings obtained as above belong to the same representation class if they are equivalent, i.e. if

$$R_2 = P^{-1}R_1P$$

for a unimodular matrix P. Ideal classes and representation classes correspond one-to-one.

Indeed, different bases of the same ideal give equivalent matrix rings, as just shown. Isomorphic ideals also give the same representation class: if $b_1, \ldots, b_n$ correspond to a basis $a_1, \ldots, a_n$, then ya_i and yb_i correspond and the same matrices occur. Conversely, if the matrix rings are equivalent, replace the basis of $\mathfrak{b}$ by the transformed basis. Then the matrices coincide, and the correspondence sending

$$c_1a_1 + \cdots + c_na_n \quad \text{to} \quad c_1b_1 + \cdots + c_nb_n$$

is an R-module isomorphism. Thus the ideals are in the same class. The class of the unit ideal is called the principal class; every basis of R gives the principal representation class.

Let C be the matrix corresponding to the element y in the representation class determined by $\mathfrak{a}$. The polynomial

$$|tE - C|$$

is invariant under equivalence of representations. Hence its coefficients are invariants of the ideal class. In particular, the coefficient of $-t^{n-1}$ is called the trace of y with respect to the class $\mathfrak{a}$, written

$$S_{\mathfrak{a}}(y) = c_{11} + c_{22} + \cdots + c_{nn}.$$

The constant coefficient, up to sign, is the norm of y with respect to the class.

For a fixed class $(\mathfrak{a})$, the traces $S_{\mathfrak{a}}(y)$ form a Z-module to which R is homomorphic as a Z-module. This follows from the matrix homomorphism:

$$S_{\mathfrak{a}}(x - y) = S_{\mathfrak{a}}(x) - S_{\mathfrak{a}}(y), \qquad S_{\mathfrak{a}}(cy) = cS_{\mathfrak{a}}(y).$$

If R has no zero divisors, the trace and norm of an element are independent of the class. For every non-zero ideal has the same rank as R, and a basis of an ideal is at the same time a basis of the unit ideal in the quotient field. Thus all traces and norms are generated in the quotient field by the unit ideal and are independent of the class.

Now let $\omega_1, \ldots, \omega_n$ be a Z-module basis of an ideal $\mathfrak{r}$. The determinant

$$|S_{\mathfrak{a}}(\omega_i \omega_j)|$$

is well defined only up to multiplication by the square of a unit of Z, but the principal ideal it generates is an invariant of the ideal $\mathfrak{r}$ and the class $(\mathfrak{a})$. A change of basis transforms the matrix cogrediently; the determinant therefore changes by the square of the transformation determinant. This principal ideal is called the discriminant ideal of $\mathfrak{r}$ with respect to the class $(\mathfrak{a})$. If R has no zero divisors, the discriminant ideal of an ideal is independent of the chosen class, because each entry in the determinant is already independent of the class.

§5. Direct sums of classes

If the ideal $\mathfrak{a}$ is a direct sum,

$$\mathfrak{a} = \mathfrak{b} + \mathfrak{c},$$

then every ideal in the class of $\mathfrak{a}$ is likewise a direct sum, with components isomorphic to $\mathfrak{b}$ and $\mathfrak{c}$ respectively. Thus one may speak of the direct sum of the ideal class and of its component classes.

Similarly, if a ring $R_{\mathfrak{a}}$ in a representation class is a direct sum of subrings which are at the same time ideals in $R_{\mathfrak{a}}$, then the same is true for every equivalent representative, and the components are equivalent rings. Thus one may speak of the direct sum of the representation class and of its component classes. The direct sum of ideal classes corresponds to the direct sum of representation classes; in this sense we simply speak of the direct sum of a class.

Let $b_1, \dots, b_s$ be a basis of $\mathfrak{b}$ and $c_1, \dots, c_t$ a basis of $\mathfrak{c}$. Since the sum is direct, the two bases together form a basis of $\mathfrak{a}$. With respect to this basis, the matrix associated with an arbitrary element of R has block-diagonal form

$$\begin{pmatrix} D^{(b)} & 0 \\ 0 & D^{(c)} \end{pmatrix}.$$

The systems of matrices with only one block non-zero form subrings, and in fact ideals, whose direct sum is the whole representation ring. They are isomorphic to the representation rings obtained from $\mathfrak{b}$ and $\mathfrak{c}$ themselves.

The trace of an element with respect to a class is the sum of the traces with respect to the component classes. If $\beta \in \mathfrak{b}$, then the trace of β with respect to $(\mathfrak{a})$ is equal to the trace of β with respect to $(\mathfrak{b})$. Hence the discriminant ideal of the component ideal $\mathfrak{b}$, taken with respect to $(\mathfrak{a})$, is equal to its discriminant ideal taken with respect to the component class $(\mathfrak{b})$.

Finally, the discriminant ideal of $\mathfrak{a}$ with respect to $(\mathfrak{a})$ is the product of the discriminant ideals of the components with respect to their component classes. This follows by taking the basis corresponding to the direct sum: the discriminant determinant becomes block diagonal, and its determinant is the product of the component determinants.

Now let X be an extension ring without zero divisors over Z such that the intersection of R and X is Z, and let

$$\bar{R} = R[X]$$

be the extension ring of the same rank as in §1, 2. If $\mathfrak{a}$ and $\mathfrak{r}$ are ideals of R, and $\bar{\mathfrak{a}}$ and $\bar{\mathfrak{r}}$ their extensions in $\bar{R}$, then the discriminant ideal of $\bar{\mathfrak{r}}$ with respect to $(\bar{\mathfrak{a}})$ is the extension ideal in X of the discriminant ideal of $\mathfrak{r}$ with respect to $(\mathfrak{a})$. One can use the same bases in the discriminant construction. In particular, the discriminant ideal of $\bar{R}$ with respect to the principal class is the extension of the discriminant ideal of R with respect to the principal class. This ideal, in X or Z, is called simply the discriminant ideal of the ring.

§6. Discriminant criterion for complete reducibility of the first kind

From now on R is again an extension ring of a field P, and $\bar{P}$ is an extension field of P. The non-vanishing of the discriminant will be a necessary and sufficient condition for complete reducibility of the first kind. By §5, 3, the discriminant ideal of R may be determined after passing to the extension ring $\bar{R} = R[\bar{P}]$. By §5, 2, it is enough to restrict attention to the primary rings out of which $\bar{R}$ is additively composed. We therefore begin with the discriminant ideals of primary rings.

If $\mathfrak{a}$ and $\mathfrak{r}$ are ideals of R and $\mathfrak{a}$ is divisible by $\mathfrak{r}$, then every basis of $\mathfrak{a}$ can be completed by adjoining elements to a linearly independent module basis of $\mathfrak{r}$. This is an immediate consequence of finite rank over the field P.

Assume R is a primary ring; let $\mathfrak{p}$ be the prime ideal belonging to the zero ideal, and let ρ be its exponent, so

$$\mathfrak{p}^{\rho} = (0), \qquad \mathfrak{p}^{\rho-1} \neq (0).$$

A special P-basis of R is obtained by successively completing a basis of $\mathfrak{p}^{\rho-1}$ to a basis of $\mathfrak{p}^{\rho-2}$, then a basis of $\mathfrak{p}^{\rho-2}$ to one of $\mathfrak{p}^{\rho-3}$, and so on, counting R as $\mathfrak{p}^0$. With such a basis one proves:

If R is a primary ring, the trace, with respect to the principal class, of every element divisible by the associated prime ideal $\mathfrak{p}$ is zero.

Indeed, with the special basis the basis elements are arranged in layers, where elements in the layer $\mathfrak{p}^i$ are not all in $\mathfrak{p}^{i+1}$. Multiplication by an element of $\mathfrak{p}$ sends each layer into the next; the corresponding matrix has zeros on the diagonal. Its trace therefore vanishes.

If the zero ideal of R is already a prime ideal and P is algebraically closed, then R has rank one and is identical with P. The unit element e may be used as basis, and

$$S(e^2) = S(e) = e.$$

The discriminant ideal is therefore the unit ideal.

On the other hand, the discriminant ideal of a proper primary ring with algebraically closed subfield P is the zero ideal. Use the special basis just described, or simply complete any basis of $\mathfrak{p}$ to a basis of R. Since the residue class ring $R/\mathfrak{p}$ has rank one, the first basis element may be taken to be the unit element e. All basis products other than e^2 are then divisible by $\mathfrak{p}$, and their traces vanish. The discriminant determinant, having at least two rows because R is properly primary, therefore has all entries except possibly $S(e^2)$ equal to zero and consequently vanishes.

Theorem. A ring R of finite rank over a subfield P is completely reducible of the first kind if and only if its discriminant ideal is the unit ideal in P, equivalently if its discriminant, determined up to the square of a unit, is non-zero.

Complete reducibility of the first kind is, by §3, 2, equivalent to complete reducibility of

$$\bar{R} = R[\bar{P}]$$

with algebraically closed $\bar{P}$. By §5, 3 the discriminant ideals of R and $\bar{R}$ are simultaneously zero or unit ideals. By §5, 2 the discriminant ideal of $\bar{R}$ is the product of the discriminant ideals of its direct-sum components, each taken with respect to its component class. But those component discriminants are exactly of the type just considered. If $\bar{R}$ is completely reducible, all components are fields isomorphic to $\bar{P}$ and all component discriminant ideals are unit ideals. Their product is a unit ideal. If $\bar{R}$ is not completely reducible, at least one component is properly primary and its discriminant ideal is zero; the product is then zero. The theorem follows.

When complete reducibility of the first kind holds, trace, norm, and discriminant admit the usual representation by conjugate elements. Let

$$R[Q] = Qe_1 + \cdots + Qe_n$$

be a decomposition as a direct sum of rank-one rings over a suitable finite extension field Q of P, chosen minimally. The components of R are represented in subfields K_i of Q, and R maps homomorphically to each K_i. With respect to the basis $e_1, \ldots, e_n$, an element

$$y = c_1e_1 + \cdots + c_ne_n$$

has diagonal matrix with entries c_i. Hence

$$S(y) = c_1 + \cdots + c_n, \qquad N(y) = c_1c_2 \cdots c_n.$$

If $a_1, \ldots, a_n$ is a module basis of R and

$$a_j = a_{1j}e_1 + \cdots + a_{nj}e_n,$$

then the discriminant is

$$|S(a_ia_j)| = \begin{vmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{vmatrix}^2 .$$

If R itself is a field, i.e. an extension field of the first kind of P, then the homomorphisms $R \to K_i$ are isomorphisms fixing P. Thus the K_i are the n conjugate fields in a common Galois extension field, each isomorphic to R. The formulas for trace, norm, and discriminant become the familiar formulas in terms of conjugates. It should again be stressed that R itself is assumed only equivalent to the K_i and need not be a subfield of the Galois extension field Q.

§7. The discriminant theorem for orders of an algebraic number or function field

The number and function fields in question - including algebraic functions of more than one indeterminate and integral algebraic functions, where one is extending a functional domain - all fit the following type of field with a prescribed notion of integral element.

Let $\mathfrak{S}$ be a principal ideal ring without zero divisors and with unit element: for example, the ring of rational integers, a polynomial ring in one indeterminate over a field, or a functional domain. Let K be its quotient field. Let L be a finite extension of the first kind of K, and let $\mathfrak{G}$ be the ring of all elements of L integral with respect to $\mathfrak{S}$. Every subring T of $\mathfrak{G}$ which contains $\mathfrak{S}$ is called an order; $\mathfrak{G}$ itself is the principal order.

Every $\mathfrak{S}$-module in $\mathfrak{G}$, in particular every ideal of an order and the order itself, has a linearly independent module basis over $\mathfrak{S}$. If L has degree n over K, it is enough to consider orders of rank

n over $\mathfrak{S}$: an order of smaller rank would, by forming quotients, generate an intermediate field of that smaller degree, and all hypotheses would be preserved there.

Thus every order T is a ring satisfying the more general hypotheses of §§4 and 5, and is in fact a ring without zero divisors. Consequently the discriminant D_T of the order is defined uniquely up to multiplication by the square of a unit of $\mathfrak{S}$. This discriminant agrees, up to a unit of $\mathfrak{S}$, with the discriminant of the field L considered as a K-module, and is therefore non-zero because L/K is an extension of the first kind. It also admits the representation in §6, 4 as the square of a determinant of conjugate basis elements.

The ideal theory in T is given by the following theorem. Every ideal of T different from the zero ideal can be represented uniquely as a product of finitely many pairwise relatively prime primary ideals, whose associated prime ideals have no proper divisors except the unit ideal.

Let p be a prime element of $\mathfrak{S}$, so that the principal ideal $\mathfrak{S}p$ is a prime ideal different from the zero and unit ideals. Let Tp be the corresponding principal ideal of T. The residue class ring

$$R = T/Tp$$

is a ring of rank n over a subfield P isomorphic to $\mathfrak{S}/\mathfrak{S}p$. The natural homomorphism $T \to T/Tp$ is always defined. The fact that R contains a subfield isomorphic to $\mathfrak{S}/\mathfrak{S}p$ follows from the second isomorphism theorem:

$$(\mathfrak{S}, Tp)/Tp \simeq \mathfrak{S}/[\mathfrak{S}, Tp] = \mathfrak{S}/\mathfrak{S}p.$$

Moreover R has rank n over P. If

$$\alpha_1, \ldots, \alpha_n$$

is a linearly independent module basis of T over $\mathfrak{S}$, and (α_i) denotes the residue class of α_i modulo Tp, then the (α_i) are linearly independent over P. For a linear relation over P would lift to a congruence

$$c_1\alpha_1 + \cdots + c_n\alpha_n \equiv 0 \pmod{Tp},$$

i.e. to

$$c_1\alpha_1 + \cdots + c_n\alpha_n = p\gamma,$$

with $\gamma \in T$. Expressing γ in the basis α_i and using uniqueness of the representation shows that every c_i is divisible by p, hence every class (c_i) vanishes in P.

Under the homomorphism the discriminant D_T passes to the discriminant of R, and the ideal decomposition of Tp passes to the decomposition of the zero ideal of R. The trace used here is the one defined by matrix representation, not the representation by conjugate elements, which is valid for T but not necessarily for R. The trace and hence the discriminant are therefore built from ring operations and ring elements and pass through the homomorphism.

If

$$Tp = \mathfrak{q}_1 \cdots \mathfrak{q}_r = [\mathfrak{q}_1, \ldots, \mathfrak{q}_r]$$

is the decomposition of Tp in T, then it passes to

$$(0) = [(\mathfrak{q}_1), \ldots, (\mathfrak{q}_r)]$$

in R. By the first isomorphism theorem,

$$R/(\mathfrak{q}_i) \simeq T/\mathfrak{q}_i,$$

so that $\mathfrak{q}_i$ and $(\mathfrak{q}_i)$ are simultaneously proper primary ideals or prime ideals. The components remain pairwise relatively prime under the homomorphism.

Discriminant theorem. A prime element p of $\mathfrak{S}$ divides the discriminant D_T of an order T if and only if, in the decomposition of the principal ideal Tp into pairwise relatively prime primary components in T, at least one proper primary component or at least one prime ideal of the second kind occurs.

If the residue class field $\mathfrak{S}/\mathfrak{S}p$ is perfect, then precisely those prime elements p which divide D_T have at least one proper primary component. If $\mathfrak{S}/\mathfrak{S}p$ is perfect and T is the principal order, then p divides the discriminant if and only if Tp is divisible at least by the square of a prime ideal of the principal order.

Indeed, by the correspondence between the ideal decomposition of Tp in T and the zero-ideal decomposition in

$$R = T/Tp,$$

the occurrence of a proper primary component or of a prime ideal of the second kind means exactly that R is not completely reducible of the first kind. By §6, 3 this is the case if and only if the discriminant of R vanishes. By the homomorphic passage just described, this is exactly the divisibility of D_T by p.

§8. The discriminant theorem for orders of relative fields

If, instead of the principal ideal ring $\mathfrak{S}$, one starts with the principal order of a number or function field - more generally, with a multiplication ring, i.e. a ring in which the ideal theory of the principal order holds - then the discriminant D_T is replaced by the discriminant ideal of T with respect to $\mathfrak{S}$. The discriminant theorem transfers completely by the following considerations.

Let $\mathfrak{S}$ be a multiplication ring: a ring without zero divisors and with unit element, in which every ideal different from the zero and unit ideals is uniquely representable as a product of powers of prime ideals, and where these prime ideals have no proper divisors except the unit ideal. Let $K, L, \mathfrak{G}, T$ be defined as above. Then the ideal theory stated in §7, 1 holds in T.

In particular, if a multiplication ring $\mathfrak{S}$ has only one prime ideal different from the zero and unit ideals, then $\mathfrak{S}$ is a principal ideal ring. For let this prime ideal be $\mathfrak{p}$, and choose $p \in \mathfrak{p}$ with $p \not\equiv 0$

(mod $\mathfrak{p}^2$). Since every ideal different from zero and unit is a power of $\mathfrak{p}$, the principal ideal $\mathfrak{S}p$ must be equal to $\mathfrak{p}$. Hence every power of $\mathfrak{p}$ is principal, and the zero and unit ideals are principal as well.

The trace of an element $\gamma \in T$ is defined as the trace of γ from L, where L is considered as a module over K. Any system of n elements of L linearly independent over K may be used as a basis to generate a matrix representation defining the trace. If $\alpha_1, \ldots, \alpha_n$ is such a system, then because L/K is an extension of the first kind,

$$|S(\alpha_i \alpha_j)|$$

is a non-zero element of K, namely the square of the determinant of the conjugate basis elements. From integral closedness of $\mathfrak{S}$ in K it follows that this determinant lies in $\mathfrak{S}$ whenever all α_i lie in $\mathfrak{G}$.

The discriminant ideal is defined as follows. Let $\tau_1, \ldots, \tau_n$ run through all systems of n elements of T, and form the determinant

$$|S(\tau_i \tau_j)|.$$

This determinant is an element of $\mathfrak{S}$, and is non-zero when the τ_i are linearly independent. The ideal of $\mathfrak{S}$ generated by all these determinants is called the discriminant ideal of the order T with respect to $\mathfrak{S}$; it is different from the zero ideal.

If

$$\beta_1, \ldots, \beta_m, \qquad m \geq n,$$

is a module basis of T over $\mathfrak{S}$, then the finitely many determinants corresponding to all choices of n of the β_i form a basis of the discriminant ideal. This follows from the module homomorphy of T to the system of traces from T: the determinants $|S(\tau_i \tau_j)|$ are linear combinations, with coefficients in $\mathfrak{S}$, of the finitely many determinants $|S(\beta_i \beta_j)|$ obtained by choosing n basis elements. If the order has a linearly independent module basis $\alpha_1, \ldots, \alpha_n$, then $|S(\alpha_i \alpha_j)|$ itself may be taken as a basis of the discriminant ideal.

We recall the passage to the quotient ring. If $\mathfrak{a}$ is an ideal of $\mathfrak{S}$, the quotient ring $\mathfrak{S}_{\mathfrak{a}}$ is the set of elements of the quotient field of $\mathfrak{S}$ having in the denominator an element prime to $\mathfrak{a}$. This quotient ring has no prime ideals other than zero, unit, and the extension ideals of the prime ideals belonging to $\mathfrak{a}$. There is a one-to-one correspondence, together with isomorphism of residue class rings, between the ideals of $\mathfrak{S}_{\mathfrak{a}}$ different from zero and unit and those ideals of $\mathfrak{S}$ whose associated prime ideals are among those belonging to $\mathfrak{a}$; zero and unit ideals correspond as well.

The same statements hold for a multiplication ring. If $\mathfrak{p}$ is a prime ideal, then $\mathfrak{S}_{\mathfrak{p}}$ has only one prime ideal different from zero and unit; hence, by the preceding paragraph, $\mathfrak{S}_{\mathfrak{p}}$ is a principal ideal ring. A generator p of the prime ideal derived from $\mathfrak{p}$ in $\mathfrak{S}_{\mathfrak{p}}$ is a prime element. The extension ideal of an arbitrary ideal $\mathfrak{c}$ of $\mathfrak{S}$ is generated by the primary component of $\mathfrak{c}$ belonging to $\mathfrak{p}$; hence it is $(p)^\rho$ or the unit ideal according as $\mathfrak{p}$ divides $\mathfrak{c}$ to exponent ρ or does not divide it. Therefore if two ideals have the same extensions in all quotient rings $\mathfrak{S}_{\mathfrak{p}}$, they are equal: they are divisible by the same prime ideals to the same powers.

Now let $\mathfrak{p}$ be a prime ideal of $\mathfrak{S}$, and write

$$B = T\mathfrak{p}$$

for its extension in T. The quotient ring T_B has a module basis of linearly independent elements over the principal ideal ring $\mathfrak{S}_\mathfrak{p}$; it is at the same time a finite $\mathfrak{S}_\mathfrak{p}$-order in the extension field L of the quotient field K of $\mathfrak{S}_\mathfrak{p}$. Therefore the discriminant theorem of §7, 3 applies to T_B over $\mathfrak{S}_\mathfrak{p}$.

The discriminant D_{T_B} generates the extension in $\mathfrak{S}_\mathfrak{p}$ of the discriminant ideal of T over $\mathfrak{S}$. Indeed, one may choose a module basis of T_B over $\mathfrak{S}_\mathfrak{p}$ consisting of elements $\alpha_1, \ldots, \alpha_n$ of T. Then $|S(\alpha_i \alpha_j)|$ belongs to the discriminant ideal, and every determinant $|S(\tau_i \tau_j)|$ in $\mathfrak{S}_\mathfrak{p}$ is divisible by it.

We finally obtain the general discriminant theorem.

General discriminant theorem. A prime ideal $\mathfrak{p}$ of a multiplication ring $\mathfrak{S}$ divides the discriminant ideal of an order T over $\mathfrak{S}$ if and only if, in the decomposition of $T\mathfrak{p}$ into pairwise relatively prime primary components in T, at least one proper primary component or at least one prime ideal of the second kind occurs.

By the preceding paragraphs, $\mathfrak{p}$ divides the discriminant ideal of T if and only if the corresponding prime element p of $\mathfrak{S}_\mathfrak{p}$ divides the discriminant of T_B. By §7, this is equivalent to the failure of complete reducibility of the first kind of the residue class ring

$$T_B/T_B p.$$

Since this residue class ring is isomorphic to

$$T/T\mathfrak{p},$$

the general discriminant theorem follows.

As a specialization to relative discriminants of number fields one obtains: a prime ideal $\mathfrak{p}$ of the principal order of a number field divides the relative discriminant of an extension field if and only if, in the principal order of the extension field, $\mathfrak{p}$ is divisible by the square of a prime ideal.

Göttingen, 30 March 1926.

With R. Brauer: On Minimal Splitting Fields of Irreducible Representations

The question of the number fields of smallest degree over a ground field P in which an irreducible representation of a finite group over P decomposes into absolutely irreducible constituents was first treated by I. Schur. Suppose, for example, that P contains the character of such a constituent. Schur showed that the degree of these fields over P - the **index** - agrees with the number of those absolutely irreducible constituents, all of which belong to the same class. Consequently the same field is already determined by the requirement that one absolutely irreducible constituent be split off. He also showed that the degree over P of every field in which such a splitting takes place is a multiple of the index.

All such fields will be called **splitting fields**, even when P does not contain the character. Schur showed by example that splitting fields of smallest degree need not be isomorphic. If P contains no associated simple character, the splitting fields must contain the field of such a character and its conjugates over P. The absolutely irreducible representations belonging to different conjugate characters are themselves conjugate, but they plainly belong to different classes. Even in that case, in order to split off a system of absolutely irreducible representations belonging to one class, it suffices to take the field of the corresponding character together with one of the corresponding splitting fields.

We now continue the study of splitting fields. Splitting fields of smallest degree can be characterized by associated division algebras; general splitting fields are characterized by two-sided simple rings invariantly connected with these division algebras. In addition, the example of the quaternion algebra shows that the degrees of minimal splitting fields are not bounded. Here a splitting field is called **minimal** if none of its proper subfields is a splitting field.

This unboundedness rests essentially on a number-theoretic existence theorem proved by H. Hasse in the immediately following note. The example is also treated elementarily and computationally. In this way, without using the general theory or the number-theoretic existence theorem, one still obtains the unboundedness of the degree numbers from a weaker but sufficient existence theorem. It should also be noted that this example again concerns splitting fields of a finite group, namely the quaternion group, which also underlies Schur's example. The representations written there are simultaneously isomorphic representations of the quaternion algebra.

1. Characterization of splitting fields

We first recall the basic concepts. Let $\mathfrak{S}$ be a hypercomplex system over a field P_0. A representation Π of $\mathfrak{S}$ over P means a system of matrices with entries in P such that Π is a homomorphic image of $\mathfrak{S}$, and such that elements p_0 of P_0 correspond to diagonal matrices p_0E. Thus P must contain

P_0. A representation Π, together with all its transforms

$$P^{-1}\Pi P,$$

forms a representation class.

The representations of finite groups are included in this definition. The associated hypercomplex system $\mathfrak{S}$, the group ring, consists of all group numbers, that is, all linear combinations of the group elements with coefficients in P_0, the original group elements being considered linearly independent and multiplication being defined by group multiplication and the laws of arithmetic. The terms reducible, irreducible, completely reducible, and absolutely irreducible are used in the usual sense; they will be applied also to representation classes. A system $\mathfrak{S}$ is called completely reducible over P if all its representation classes over P are completely reducible.

Assume now that $\mathfrak{S}$ is completely reducible over P_0, and that P_0 is a perfect field. Then the same is true for every algebraic extension P of P_0, and the representation classes remain completely reducible after every such extension. The center of $\mathfrak{S}$ is a direct sum of finitely many fields. Every extension P of P_0 isomorphic to one of these fields is the field of a simple character.

If P_0 is extended to such a field P, a two-sided simple ring splits off from $\mathfrak{S}$; to it belongs the absolutely irreducible representation class of the character. By Wedderburn's theorem this ring is the direct product of a matrix ring over P with a division algebra A, determined uniquely up to isomorphism. The center of A is P, and A has degree m^2 over P, where m is Schur's index attached to the character.

Every splitting field of $\mathfrak{S}$, with respect to the representation belonging to that character, is at the same time a splitting field of A with respect to the representations over P, and conversely. The conjugate representations of $\mathfrak{S}$ are obtained by starting with the corresponding two-sided simple ring over P_0; the associated division algebra is then isomorphic to A and has the conjugate field as its center. Thus the problem of splitting fields is completely reduced to the associated division algebra A and to its splittings.

In particular, the following theorem holds.

All maximal commutative subfields of A have the same degree m over P, where m is Schur's index. All these maximal commutative subfields are splitting fields of smallest degree, and every splitting field of smallest degree is contained in one of them. The degree of every splitting field is a multiple of m. The algebra A has only one absolutely irreducible representation class, and likewise only one irreducible representation class over P.

To characterize general splitting fields, let A_r denote the direct product of A with the full matrix ring of degree r over P. Then A_r is two-sided simple and has A as its associated division algebra. Conversely, every two-sided simple hypercomplex ring over P to which A is associated is obtained in this way. Equivalently, A_r may be characterized as the ring of all r by r matrices with entries in A.

Every splitting field of A of degree mr, if such a field exists, is isomorphic to a maximal commutative subfield of A_r. Conversely, all maximal commutative subfields of A_r are splitting fields, each of degree ms, where s divides r. In the regular case all maximal commutative subfields of A_r have degree mr. The regular case means that the ground field P has the following property: for every commutative field Z over P, finite over P, there exist commutative extensions of Z of arbitrarily prescribed degree.

Whether, for $r > 1$, minimal fields occur among these splitting fields is not decided merely by the embedding in A_r. The following example shows that this can in fact happen for infinitely many r.

2. Unboundedness of the degrees of minimal splitting fields

For this unboundedness we use the quaternion algebra as example, regarded as a hypercomplex system over the rational field P. It has basis elements i, j, k, besides the unit 1, with the familiar relations

$$i^2 = j^2 = k^2 = -1, \qquad ij = k, \qquad jk = i, \qquad ki = j,$$

with the reversed products changing sign.

We shall show that the splitting fields are exactly those algebraic number fields in which, after adjoining the field, there exists an idempotent element r, i.e.

$$r^2 = r,$$

with r neither zero nor one. These number fields can also be characterized as those in which -1 is representable as a sum of three squares, and consequently as a sum of two squares. Indeed, write

$$r = a + \frac{\beta}{2}i + \frac{\gamma}{2}j + \frac{\delta}{2}k.$$

Then

$$r^2 = r$$

is equivalent to the system

$$4a^2 - 4a - \beta^2 - \gamma^2 - \delta^2 = 0, \qquad 2a\beta - \beta = 0,$$

$$2a\gamma - \gamma = 0, \qquad 2a\delta - \delta = 0.$$

Since β, γ, δ are not all to vanish, this is equivalent to

$$-1 = \beta^2 + \gamma^2 + \delta^2.$$

The fact that representability of -1 as a sum of three squares implies representability as a sum of two squares follows from the identity

$$(c^2 + d^2)(a^2 + b^2 + c^2 + d^2) = (ac + bd)^2 + (ad - bc)^2 + (c^2 + d^2)^2,$$

which is also a consequence of the product theorem for quaternion norms.

Every such idempotent element produces a splitting, and the existence of such an idempotent is necessary for splitting. This follows from standard facts about completely reducible systems. Every irreducible representation of a completely reducible system is generated by a one-sided simple ideal, more precisely by the corresponding ideal class, and every such ideal class generates the representation. These one-sided simple ideals all occur in one-sided direct decompositions, and those decompositions determine the primitive idempotent elements, the components of the identity. Conversely, every idempotent element gives a decomposition

$$\mathfrak{S} = r\mathfrak{S} + (1-r)\mathfrak{S};$$

primitive idempotents split off simple ideals.

A division algebra, regarded as a hypercomplex system over its center and then extended to a splitting field, becomes a direct sum of m absolutely simple one-sided ideals, where m is Schur's index. For the quaternion algebra $m = 2$. Hence every idempotent different from 0 and 1 already gives the decomposition into absolutely simple ideals, and this corresponds to the splitting into absolutely irreducible representation classes. Thus the fields described above are exactly the splitting fields.

It follows further that all cyclic fields K_n of degree 2^n over the rational field in which -1 is representable as a sum of two squares are minimal splitting fields of the quaternion algebra. Such a splitting field cannot be real, because -1 is a sum of squares in it. On the other hand, the subfield of degree 2^{n-1} of K_n, and hence every proper subfield of K_n, is necessarily real: it is the intersection of K_n with the field of all real algebraic numbers. Thus K_n is Galois of degree two over this intersection, which must be the subfield of degree 2^{n-1}.

By Hasse's existence theorem there are such fields K_n for every n; hence the degrees of the minimal splitting fields are not bounded. In fact every power of the index occurs here as the degree of a minimal splitting field.

3. Elementary treatment of the splitting fields of the quaternion algebra

Let $\mathfrak{S}$ denote the representation of the quaternion group generated by the usual two by two matrices I, J, K over a field. The group ring generated by $\mathfrak{S}$ is just a representation of the quaternion algebra as a hypercomplex system. We give an elementary proof that $\mathfrak{S}$ is representable over a field K if and only if -1 is representable in K as a sum of two squares.

Suppose first that

$$\mathfrak{S}^* = T^{-1}\mathfrak{S}T$$

is a representation rational over K. Write

$$I^* = T^{-1}IT, \qquad J^* = T^{-1}JT.$$

Since J and J^* are similar matrices rational over K, there is a transformation rational over K carrying J^* to J. Replacing $\mathfrak{S}^*$ by an equivalent K-rational representation, we may therefore assume $J^* = J$. Let

$$I^* = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

with $a, b, c, d \in K$. From

$$IJ = -JI$$

one obtains $I^*J = -JI^*$, hence $a = -d$ and $b = c$. From $I^2 = -E$ follows

$$(I^*)^2 = -E,$$

which gives

$$a^2 + b^2 = -1.$$

Thus -1 is indeed a sum of two squares in K.

Conversely, suppose that in K one has

$$-1 = \alpha^2 + \beta^2.$$

Set

$$I^* = \begin{pmatrix} \alpha & \beta \\ \beta & -\alpha \end{pmatrix}, \qquad J^* = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

and put $K^* = I^*J^*$. A direct calculation shows that

$$\pm E, \quad \pm I^*, \quad \pm J^*, \quad \pm K^*$$

form a group rational over K similar to the quaternion representation. This proves the characterization.

It remains to show, elementarily, that for infinitely many n there are cyclic fields of degree 2^n which are minimal splitting fields of the quaternion group. Let p be a rational prime for which, for some $r > 0$,

$$2^r + 1 \equiv 0 \pmod{p},$$

and let 2^n be the highest power of 2 dividing $p - 1$. There are primes p belonging to arbitrarily large values of n. For if k is any positive integer and p is a prime divisor of $2^{2^k} + 1$, then the congruence just required holds with $r = 2^k$. Moreover 2 modulo p has exponent 2^{k+1}, and hence the highest power 2^n dividing $p - 1$ has $n > k$.

Let ε be a primitive p-th root of unity. We show that in $P(\varepsilon)$, where P is the rational field, one can represent -1 as a sum of two squares:

$$-1 = a^2 + b^2 = (a + ib)(a - ib), \qquad a, b \in P(\varepsilon).$$

This is equivalent to saying that in the field of $4p$-th roots of unity,

$$P(\varepsilon, i) = P(\varepsilon_{4p}),$$

there is an element whose relative norm over $P(\varepsilon)$ is exactly -1.

The element $1 + i\varepsilon^{\nu}$ has relative norm

$$(1 + i\varepsilon^{\nu})(1 - i\varepsilon^{\nu}) = 1 + \varepsilon^{2\nu}.$$

Thus all numbers $1 + \varepsilon^{2\nu}$ are relative norms from $P(\varepsilon, i)$ over $P(\varepsilon)$. Hence so is their product

$$\prod_{u=0}^{r-1} (1 + \varepsilon^{2\nu}).$$

Using the congruence $2^r \equiv -1 \pmod{p}$, this product becomes -1. Also ε itself is a relative norm. Hence -1 is a relative norm, and so -1 is representable in $P(\varepsilon)$ as a sum of two squares. Thus $P(\varepsilon)$ is a splitting field for the quaternion group.

Let K be the cyclic subfield of $P(\varepsilon)$ of degree 2^n. We claim that K is itself a splitting field. Let m be the index of the quaternion algebra with respect to K as ground field. Then $m = 1$ or $m = 2$. But $P(\varepsilon)$ is a splitting field of odd relative degree over K, so the case $m = 2$ is impossible. Therefore $m = 1$, which means that K is a splitting field.

Thus for infinitely many n there exist cyclic fields of degree 2^n which are minimal splitting fields. # Hypercomplex quantities and representation theory in arithmetical conception

Introduction

The most important general theorems on hypercomplex systems go back to Molien. Almost independently, Frobenius soon afterwards developed the theory of hypercomplex systems and their representations, especially the representation theory of finite groups, in a unified way. The basis is Dedekind's notion of the group determinant, more generally the determinant of an arbitrary hypercomplex system.

Frobenius shows that the different irreducible factors of this determinant, with the complex numbers as coefficient field, correspond to the different irreducible representation classes, and that all irreducible representation classes are exhausted in this way. Such a representation class is completely characterized by its character system. For finite groups, this character system arises by decomposing the determinant of the commutative hypercomplex system obtained from the conjugacy classes of the group. That determinant decomposes into linear factors whose coefficients are the characters. This is a direct generalization of Dedekind's result that the group determinant of a finite abelian group decomposes into linear factors whose coefficients are the several characters of the abelian group.

Two questions are separated. The first is the reduction and decomposition of a given representation of a given ring over a given, not necessarily commutative, representation field. The second is the determination of all possible representations, or even only of all irreducible representations, of the ring in that field.

The first question is the simpler one. No special assumptions about the ring to be represented, or about the possibly non-commutative representation field, are needed. The fact that one deals with matrices of a fixed degree already supplies the necessary finiteness. The question can be treated completely by means of the theory of groups with operators.

A group M is said to have an operator domain if, together with the group, there is given a system of symbols

$$O, H, \dots$$

so that $O(a), H(a), \dots$ are defined uniquely for every $a \in M$, again as elements of M, and so that the distributive law

$$O(ab) = O(a)O(b)$$

holds. Two groups with the same operator domain are operator-isomorphic if they are isomorphic as ordinary groups and if, whenever a corresponds to $\bar{a}$, the elements $O(a)$ and $O(\bar{a})$, $H(a)$ and $H(\bar{a})$, and so on, correspond as well.

Only subgroups which themselves admit the operator domain are allowed. Thus a group is simple if it has no proper admissible normal subgroup except the identity. With this convention the Jordan-Hoelder theorem remains valid: if a group with operators has a composition series at all, the system of factor groups is uniquely determined, up to order, in the sense of operator-isomorphism. Under the same hypothesis there is also a direct-product theorem: the decomposition into directly indecomposable factors is unique in the same sense.

The connection with representation theory is that ideals and modules are examples of groups with operators. They are abelian groups with respect to addition, restricted by admitting specified multiplications by elements of a ring. Every representation is generated by a representation module, that is, by a module with respect to the ring and the representation field. More precisely, every module class, meaning a class of mutually operator-isomorphic representation modules, generates the same representation.

The uniqueness of the reduction of a representation is therefore answered by the composition-series theorem and the direct-product theorem. Applied to the representation module, a composition series gives, for every matrix, a block reduction in which the diagonal blocks corresponding to the factor groups generate irreducible representations, in the sense that they allow no further such reduction. Since the classes of those factor groups are uniquely determined, the diagonal representations are uniquely determined up to equivalence. Similarly, the direct-product theorem gives the uniqueness of the splitting into indecomposable constituents; each constituent can then be reduced uniquely by its own composition series.

The second question is only sketched here for hypercomplex systems, and more generally for rings satisfying the double chain condition. The result is that every irreducible, or simple, module class is an ideal class. Thus composition series of one-sided ideals, through their factor groups, already supply all irreducible representations. More exactly, it is enough to take composition series in the residue class ring modulo the radical; this quotient becomes a completely reducible ring.

The representation problem is thereby reduced to the structural study of completely reducible rings. It also becomes natural to start with representations in the automorphism division rings. All simple right ideals of a two-sided simple component of such a ring belong to one class and therefore have the same automorphism ring. Because the ideals are simple, this automorphism ring is a division ring, and it is at the same time the automorphism ring of the simple left ideals. The simple ring itself becomes isomorphic to the full matrix ring over this automorphism division ring.

All representations over a subfield, and in particular over the coefficient domain of the hypercomplex system, arise from this matrix representation by replacing the elements of the automorphism division ring by the corresponding matrices. In this way Wedderburn's structure theorem for hypercomplex systems is reinterpreted through the general group-theoretic notion of the automorphism division ring, and is at the same time recognized as the source of the representation theory of hypercomplex systems.

The new questions which appear are those of a Galois theory for non-commutative division rings, closely connected with the question of splitting fields. At the same time one obtains a prospect of structure theorems for more general, not completely reducible, rings through the automorphism rings of indecomposable ideals. Group theory, extended to the most general groups with operators, again appears as the organizing principle.

Hypercomplex quantities and representation theory

Introduction

The general theory just indicated is now developed in a strictly arithmetical form. Hypercomplex quantities and representation theory are not treated as separate subjects, but as parts of one theory: the theory of module classes and ideal classes over non-commutative rings satisfying suitable finiteness conditions.

The central assertion is that the irreducible module classes are already exhausted by the corresponding ideal classes. In particular, for rings without radical, every module class decomposes into irreducible classes. This is the arithmetical counterpart of Frobenius's theorem that the irreducible factors of the regular group determinant, or more generally of the system determinant, already give all irreducible representation classes, and that for systems without radical complete reducibility holds.

The determinant and its factorization may be regarded as a passage to the norm. Here, instead, the direct-sum decomposition of ideals and their composition series are treated directly. Representation theory becomes a theory of module classes by means of the representation module.

The introduction of module and ideal classes is group-theoretic. Modules and ideals are abelian groups under addition, with additional operators supplied by multiplication by ring elements. They are therefore groups with operators. For such groups the ordinary isomorphism is replaced by operator-isomorphism; groups which are operator-isomorphic are collected into a class. In the case of ideals of a number field this class notion coincides with the usual one.

The theory of groups with operators supplies the group-theoretic foundation. Composition series, direct products or direct sums, and completely reducible groups all retain their usual uniqueness properties, interpreted in the sense of operator-isomorphism. These results will be used first to obtain the general uniqueness theorem for the irreducible diagonal constituents of arbitrary representations. Since representation classes correspond one-to-one to the classes of representation modules, and these are classes of groups with operators, the assertion follows formally from the group theory.

The further question, the determination of all representations, requires a closer analysis of the structure of the rings to be represented. In the case of hypercomplex systems this means an arithmetical reconstruction and extension of the Wedderburn results, guided by the study of one-sided ideals. It turns out that the multiple-chain condition for right ideals, or equivalently the minimal condition for right ideals, is already sufficient as the finiteness hypothesis.

Rings without radical satisfying this minimal condition are exactly the completely reducible right rings with identity. In such rings all simple right ideals of a two-sided indecomposable ring belong to one class; their automorphism ring is a division ring; and the ring itself is a full matrix ring over that division ring. From this, the representation theory and the questions about splitting fields follow in the same conceptual framework.

Chapter I. Group-theoretic foundations

§1. Groups with operators

Let G be a group, finite or infinite. An operator domain Ω for G is a set of new symbols

$$O, H, \ldots$$

such that for each $a \in G$ and each $O \in \Omega$ an element $Oa \in G$ is uniquely defined and the distributive law

$$O(ab) = Oa \cdot Ob$$

is satisfied. Thus every operator defines a homomorphism of the group into itself, possibly onto a proper subgroup. The identity is carried to the identity, and inverses to inverses.

An operator domain is called absolute if distinct operators define distinct homomorphisms. From any operator domain one obtains an absolute one by identifying all operators which define the same homomorphism. An absolute operator domain is then in one-to-one correspondence with a subset of the set of all endomorphisms of the group.

A subgroup A of G is admissible, with respect to the fixed operator domain Ω, if it is stable under the operators: for every $a \in A$ and every $O \in \Omega$ one has $Oa \in A$.

As examples, take first the inner automorphisms

$$a \mapsto c^{-1}ac.$$

The admissible subgroups are precisely the normal subgroups. If all automorphisms are taken as operators, then the admissible subgroups are the characteristic subgroups, invariant under every automorphism.

A ring is also an example. A ring is here an abelian group under addition equipped with multiplication satisfying

$$r(a+b) = ra + rb, qquad(a+b)r = ar + br, qquad(ab)c = a(bc).$$

Each element r gives two operators, left multiplication $x \mapsto rx$ and right multiplication $x \mapsto xr$. The admissible subgroups are then the ideals: left ideals if they are stable under all left multiplications, right ideals if they are stable under all right multiplications, and two-sided ideals if they are stable under both.

The ideals become only the trivial ideals when the ring is a division ring, that is, when the non-zero elements form a group under multiplication. No commutativity of multiplication is assumed here, either for rings or for division rings.

Let o be a ring, and let M be an abelian group, written additively. Suppose that a multiplication of elements of o with elements of M is given on the left, and that

$$r(a+b) = ra + rb, \qquad (rs)a = r(sa), \qquad (r+s)a = ra + sa$$

for $r, s \in o$ and $a, b \in M$. Then M is a left o-module. The ring o is at the same time an operator domain. If different elements of o induce different operations, the multiplier domain is absolute. As before, one passes from an arbitrary multiplier domain to the absolute one by identifying elements which induce the same operation; the absolute multiplier domain is again a ring. The admissible subgroups of a module are its submodules. Taking $M = o$ gives the left ideals.

Right modules are defined analogously by the rules

$$(a+b)r = ar + br, \qquad a(rs) = (ar)s, \qquad a(r+s) = ar + as.$$

If M is simultaneously a left module over o and a right module over o', and if

$$r(ar') = (ra)r'$$

for $r \in o$, $a \in M$, $r' \in o'$, then M is a bimodule.

Finally, for an abelian group M the endomorphisms of M form a ring. Addition and multiplication are defined by

$$(H+O)a = Ha + Oa, \qquad (HO)a = H(Oa).$$

With respect to this ring, the abelian group itself is a module. From now on, when subgroups are mentioned in the context of groups with operators, only admissible subgroups are meant. The intersection of two admissible subgroups is admissible. The product of two such subgroups is admissible whenever at least one of them is normal.

§2. The isomorphism theorems

Let two groups have the same operator domain. A map from G onto $\bar{G}$ is an operator-homomorphism if it is a homomorphism in the ordinary sense and, moreover, if a maps to $\bar{a}$ then Oa maps to $O\bar{a}$ for every operator O. If the correspondence is one-to-one, it is an operator-isomorphism.

The homomorphism theorem has the same form as for ordinary groups. If $\bar{G}$ is an operator-homomorphic image of G, then $\bar{G}$ is operator-isomorphic to a factor group G/N, where N is the admissible normal subgroup consisting of the elements of G which map to the identity of $\bar{G}$. Conversely, every admissible normal subgroup N defines a factor group G/N which admits the same operator domain and is an operator-homomorphic image of G.

A group is simple if it has no normal subgroups except itself and the identity group. Every homomorphism of a simple group is either an isomorphism or sends every element to the identity.

The first isomorphism theorem says: if $\bar{G}$ is a homomorphic image of G, if A is a normal subgroup of $\bar{G}$, and if B is the set of elements of G which map into A, then B is a normal subgroup and

$$\bar{G}/A \simeq G/B.$$

In the notation of the homomorphism theorem, if $\bar{G} = G/N$, then A necessarily contains the image of N, and one recovers A from its inverse image. The formula can be written as

$$(G/N)/(B/N) \simeq G/B.$$

The second isomorphism theorem says: if A is a subgroup and B is a normal subgroup of G, then $A \cap B$ is normal in A, and

$$AB/B \simeq A/(A \cap B).$$

Indeed, under the homomorphism $G \to G/B$, the elements of A go to the cosets making up AB/B, and the kernel is $A \cap B$.

An operator-homomorphic map of a group into itself, or into a subgroup of itself, will be called an endomorphism of the group. If the group is abelian and written additively, the operator-endomorphisms form a ring under the addition and multiplication just described; this ring is the automorphism ring in Noether's terminology.

The automorphism ring of a simple abelian group with operators is a division ring. For every endomorphism either maps the group onto itself or maps it to the zero group. The non-zero endomorphisms are isomorphisms, and the isomorphisms of the group onto itself form a group under multiplication. Hence the ring with the zero operator omitted is a group, which is exactly the assertion that the ring is a division ring.

If G is a left o-module and the operator-endomorphisms are written as right operators, then G becomes a bimodule over o on the left and over its automorphism ring on the right. The identities

$$(a+b)T = aT + bT, \qquad (ra)T = r(aT)$$

state precisely that T is a homomorphism with respect to the left operators. Conversely, in any bimodule the same formulas say that each right multiplier induces an operator-endomorphism relative to the left multipliers.

§3. Composition series

A finite chain of admissible subgroups

$$G = A_0 \supset A_1 \supset \cdots \supset A_r = E$$

is a composition series if each A_i is normal in the preceding subgroup and no further subgroup with the same property can be inserted between two consecutive terms. The factor groups

$$A_0/A_1, A_1/A_2, \ldots, A_{r-1}/A_r$$

are the composition factors. They are simple groups.

If all inner automorphisms are included among the operators, such a series consists entirely of normal subgroups of G and is called a principal series. If all automorphisms are included, it is called a characteristic series. In the examples above one obtains composition series of ideals, modules, or bimodules.

The Jordan-Hoelder theorem remains valid in this setting. If a group with operators has two composition series, then the two series have the same length, and their composition factors are pairwise operator-isomorphic after a suitable rearrangement. Moreover, if a composition series exists, then one can draw such a series through every admissible normal subgroup.

The existence of a composition series is automatic for finite groups, and more generally for groups satisfying both the maximal and minimal conditions. The maximal condition says that every set of admissible subgroups contains a maximal member, equivalently that every ascending chain

$$A_1 \subset A_2 \subset A_3 \subset \cdots$$

terminates. The minimal condition says that every set of admissible subgroups contains a minimal member, equivalently that every descending chain terminates.

When only normal subgroups are admissible, as for abelian groups, the existence of a composition series conversely implies both chain conditions. Each normal subgroup has a composition series, and its length is the length of the normal subgroup. If $A \subset B$, the length of A is smaller than the length of B, since a composition series for B can be drawn through A. Hence a subgroup of shortest length is minimal, and one of greatest length is maximal.

§4. Direct products and direct sums

A group G is the direct product of two factors, written

$$G = A \times B,$$

if A and B are normal subgroups, $AB = G$, and $A \cap B = E$. Equivalently, every element $g \in G$ has a unique expression

$$g = ab, \qquad a \in A,\ b \in B,$$

and the elements of A commute with the elements of B.

For finitely many factors one writes

$$G = A_1 \times \cdots \times A_n.$$

This means that, for each i, G is the direct product of A_i and the product of all remaining factors. Equivalently, every element has a unique representation

$$g = a_1 a_2 \cdots a_n, \qquad a_i \in A_i,$$

and the elements of different factors commute with one another.

A group is directly indecomposable if it cannot be written as a direct product of two proper factors. Every group satisfying the minimal condition is a direct product of finitely many directly indecomposable groups. In additive notation, products become sums and direct products become direct sums.

The direct-product theorem used later is a uniqueness theorem in the sense of operator-isomorphism. Under the chain hypotheses stated above, the decomposition into directly indecomposable factors is unique up to order and operator-isomorphism. This is the additive form that will be applied to ideals and modules.